

Partisan Professionals: Evidence from Credit Rating Analysts

ELISABETH KEMPF and MARGARITA TSOUTSOURA

ABSTRACT

Partisan perception affects the actions of professionals in the financial sector. Linking credit rating analysts to party affiliations from voter records, we show that analysts not affiliated with the U.S. president's party downward-adjust corporate credit ratings more frequently. Since we compare analysts with different party affiliations covering the same firm in the same quarter, differences in firm fundamentals cannot explain the results. We also find a sharp divergence in the rating actions of Democratic and Republican analysts around the 2016 presidential election. Our results show that analysts' partisan perception has price effects and may influence firms' investment policies.

RECENT EVIDENCE SUGGESTS A LARGE increase in polarization across political parties in the United States (e.g., Iyengar, Sood, and Lelkes (2012), Mason (2013), Lott and Hassett (2014), Mason (2015), Gentzkow (2016), Boxell, Gentzkow, and Shapiro (2017)). In particular, voters exhibit an increased tendency to view the economy through a “partisan perceptual screen,”¹ that

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¹ Campbell et al. (1960) introduce the idea of the partisan perceptual screen, writing that “Identification with a party raises a perceptual screen through which the individual tends to see what
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is, their assessment and interpretation of economic conditions and policies depend to a large extent on whether the White House is occupied by the party they support (e.g., Bartels (2002), Gaines et al. (2007), Gerber and Huber (2009), Curtin (2016), Mian, Sufi, and Khoshkhou (2018)).

To understand how partisan perceptions affect the U.S. economy, it is important to establish whether and when they translate into differences in economic agents' behavior. Whereas prior work documents partisan bias in households' assessment of future economic conditions, evidence on actual economic behavior is mixed.² Moreover, the extent to which partisan perception influences the economic expectations and actions of individuals with greater economic sophistication, in high-stakes professional environments, remains an open question.³

In this paper we fill these gaps by investigating how partisan perception affects the actions of an important set of professionals in the financial sector, namely, credit rating analysts. Focusing on credit analysts provides an interesting setting because their expertise and career concerns should reduce the effect of partisan perception (e.g., Gentzkow, Glaeser, and Goldin (2006), Hong and Kacperczyk (2010)). At the same time, any effect of partisan perception on credit rating actions is likely to have implications for firms' cost of financing (Fracassi, Petry, and Tate (2016)), as well as for their financial policy and investment decisions (Chernenko and Sunderam (2011), Begley (2015), Almeida et al. (2017)).

To identify the effect of partisan perception, we test whether credit analysts' rating actions depend on their political alignment with the U.S. president. This test poses a number of empirical challenges. First, it requires observable actions at the level of the individual analyst. Second, analysts need to be linked to information about their political affiliation. Third, it requires comparing the actions of analysts with different political affiliations on the same task and in the same information environment. And fourth, it requires separating the effect of political alignment with the president from time-invariant characteristics of Democratic and Republican analysts.

To address these challenges, we compile a novel hand-collected data set that links credit rating analysts to the ratings they issue, as well as to information on party affiliation from voter registration records. Our sample contains 557 corporate credit analysts with nonmissing information on their party affiliation, that work at Fitch, Moody's, or Standard and Poor's (S&P) over the period 2000 to 2018. These analysts cover a total of 1,984 U.S. firms. By comparing rating actions of analysts who rate the same firm at the same point in time,

is favorable to his partisan orientation" (Campbell et al. (1960), p. 133). In this paper, we use the terms "partisan perceptual screen," "partisan perception," and "partisan bias" interchangeably.

² Whereas Makridis (2019) documents a significant effect of partisan bias on household spending, McGrath (2017) and Mian, Sufi, and Khoshkhou (2018) find no significant effect. Focusing on households' investment decisions, Meeuwis et al. (2018) show that political affiliation affects portfolio choice around the U.S. election of November 2016.

³ Notable exceptions are Jelveh, Kogut, and Naidu (2018), who document partisan bias in economic research, and Posner (2008), McKenzie (2012), and Chen (2019), who discuss partisan bias among judges.

we ensure our results cannot be driven by differences in the fundamentals of rated firms (i.e., we can compare analysts on the same “task”).

We find that partisan perception affects credit ratings. Analysts who are not affiliated with the president’s party are more likely to adjust ratings downward, relative to other analysts covering the same firm at the same point in time. Specifically, analysts who are not affiliated with the president’s party downward-adjust ratings more by 0.013 notches per quarter. This effect corresponds to 11.4% relative to the average absolute quarterly rating adjustment and is therefore economically sizable. Over a four-year presidency (i.e., 16 quarters), these estimates imply that analysts who are misaligned with the president downward-adjust the rating of the average firm by 0.21 ($= 0.0134 \times 16$) notches more than aligned analysts. This effect corresponds to a one-notch rating downgrade (e.g., A to A–) of approximately one out of every five firms. Overall, the effect of partisan perception that we document is comparable to other nonfundamental factors influencing rating agencies’ information production identified in the literature, such as the effect of competition or the home-bias effect.

Our empirical strategy ensures that our result cannot be explained by several potential confounding factors. Most importantly, following Fracassi, Petry, and Tate (2016), we control for nonrandom matching of analysts to firms by including firm $\times$ quarter fixed effects in the regressions. This design rules out the possibility that Democratic analysts rate firms that tend to do well under the policies of Democratic presidents. Our empirical strategy also allows us to control for differences in rating methodologies across rating agencies via agency $\times$ quarter fixed effects. Finally, we control for unobserved time-invariant differences across analysts with different party affiliations using party-affiliation fixed effects. We therefore focus on changes in analysts’ behavior that vary with whether their preferred party is in power, as opposed to static differences between Democratic and Republican analysts.

To further support our conclusion that our finding reflects partisan perception, we conduct an event study around the 2016 presidential election. The 2016 election provides a particularly clean setting because the outcome was unexpected and the two candidates had very different views on economic policy. We find a sharp and sizable divergence in the rating actions of Democratic and Republican analysts (see Figure 4). To the best of our knowledge, this study is among the first to document that the highly polarized 2016 election was accompanied by a differential response in the behavior of sophisticated economic agents. Furthermore, we find substantially larger effects for analysts who are more politically active, as proxied by the frequency with which an analyst votes.

We next show that the rating actions of partisan analysts have nonnegligible price as well as real effects. Regarding price effects, we begin by documenting that the stock price response to a downgrade does not vary with whether the downgrade is announced by an analyst who is ideologically misaligned or aligned with the president. In other words, securities prices do not seem to differentiate between analysts’ ideological leanings. As a result, replacing an

analyst who is aligned with the president with an analyst who is misaligned leads to a difference in the firm's market capitalization of 0.52% to 0.62%, or \$89 million to \$107 million, over a four-year presidential term. We also find a significant increase in bond yields of 6.1 basis points (bps) over a four-year period. As we argue below, these effects likely represent lower bounds for the true price effects of analysts' partisan perception. We also show that firms rated by analysts who transition from aligned to misaligned with the president experience a significant decrease in firm investment around presidential elections.

Having established the consequences of partisan perception on credit ratings, securities prices, and firm investment, we investigate the economic mechanism. We interpret the evidence in this paper as indicating that analysts with opposing political views differ in their beliefs about how the economic policies of the U.S. president affect the credit risk of firms in the economy. One important advantage of our setting for isolating belief disagreement from other factors is that the rating actions of analysts are unlikely to be driven by how the election of their preferred candidate affects analysts' personal economic condition. To further support our interpretation, we provide three additional pieces of evidence. First, we conduct an online survey of credit rating analysts and find striking differences in the assessment of current economic conditions by Democrats and Republicans, consistent with existing evidence from U.S. households. Second, we show that analysts' alignment with the president's party has no effect on the ratings of firms with low cyclicality. This result implies that the disagreement corresponds to the set of firms whose fundamentals should be most affected by changing aggregate economic conditions. Third, we investigate whether the effect is more pronounced in periods when views of economic conditions are more politically polarized in the U.S. population. We use the absolute difference in the views of economic conditions between Democrats and Republicans from the Gallup Daily Survey to capture political polarization in economic views. We find that the effect of political alignment with the president is 83% larger when polarization increases by one standard deviation.

This study is the first to identify a significant effect of partisan perception on the actions of finance professionals, specifically, on the rating actions of credit analysts. If partisan perception affects the decisions of credit rating analysts, it may also affect decisions of other relevant economic agents. Given that the effect of partisan perception prevails in a setting where pecuniary and professional gains are at stake, it may be even more pronounced in less competitive labor markets. We look forward to future research exploring this phenomenon in other labor market settings.

The rest of this study proceeds as follows. In Section I, we discuss the related literature. Section II presents the data, sample construction, and summary statistics. Section III describes the empirical strategy. Section IV examines the extent to which analysts' rating actions are influenced by partisan perception. Section V investigates the price and real effects of partisan perception. Section VI discusses the economic mechanism. Finally Section VII concludes.

I. Motivation and Related Literature

Our study is motivated by growing evidence that partisanship has become more pervasive in the United States, with partisan conflict penetrating into a greater number of issue areas (e.g., Brewer (2005)). According to Doherty (2017), party identification is now a more significant predictor of Americans' fundamental political values than any other social or demographic divide, including gender, race, education, and religion. Similarly, Bertrand and Kamenica (2018) find that differences in social attitudes by political ideology have increased in the United States since the 1970s, whereas they do not find a similar increase in differences across gender or race. Moreover, a growing literature, described in detail below, documents the importance of political partisanship as a predictive variable for the economic expectations of U.S. households. The documented importance of partisan perception for individuals' economic views and for society more broadly highlights the need for an empirical study on how political partisanship shapes information production in financial markets. Our paper fills this gap and contributes to several strands of the literature.

First, our findings contribute to a growing literature on the connection between partisanship and economic behavior. Most existing studies focus on households, and studies of consumption behavior produce mixed results. In an early paper, Gerber and Huber (2009) demonstrate that consumption changes following a political election are correlated with whether the election was won by the respondent's preferred political party. Analyzing Australian elections, Gillitzer and Prasad (2018) find that changes in sentiment around elections are associated with future vehicle purchase rates. Benhabib and Spiegel (2018) document a positive relation between partisan-related sentiment and state-level GDP growth. Makridis (2019) uses individual-level data from Gallup and shows that self-reported consumption of nondurable goods rose more among conservatives around the 2016 presidential election. However, other studies cast doubt on a significant connection between partisanship and household consumption. McGrath (2017) extends the sample in Gerber and Huber (2009) and finds no evidence of an effect of partisan ideology on spending. Combining data on vehicle purchases and credit-card spending with an estimated propensity to vote for the Republican candidate in presidential elections at the county and state levels, Mian, Sufi, and Khoshkhoh (2018) find a significant relationship between party affiliation and economic expectations but not between party affiliation and household spending.⁴ Prior studies also examine partisanship and household asset allocation. Addoum and Kumar (2016) show that the industry-level composition of investors' portfolios changes when the party in power changes. Bonaparte, Kumar, and Page (2017) find that investors' portfolio allocation to risky assets is influenced by whether

⁴ Several factors could explain the mixed findings when linking partisan ideology to household consumption, such as using survey-based, self-reported consumption data versus administrative data, studying different countries and time periods, and employing different methods to infer political affiliation.

their preferred party is in power. Meeuwis et al. (2018) document that Republican investors actively increase the equity share and market beta of their portfolios relative to Democrats following the U.S. election of November 2016. We add to this literature by establishing that partisan perception affects the behavior of finance professionals and has nontrivial price and real effects.

Our results also contribute to studies that investigate the effect of partisan ideology among other groups of professionals. Hersh and Goldenberg (2016) find evidence of partisan bias among medical doctors, as doctors with different political affiliations recommend different treatment plans for politically sensitive health issues. Posner (2008), McKenzie (2012), and Chen (2019) document evidence of partisan bias among judges. Our work complements these studies by providing evidence on the bias of financial experts.

This study also extends the literature on nonfundamental determinants of credit ratings at the analyst level. Fracassi, Petry, and Tate (2016) find evidence of systematic optimism and pessimism among credit analysts and show that these biases affect credit spreads. Cornaggia, Cornaggia, and Israelsen (2019) and Cornaggia, Cornaggia, and Xia (2016) document that home bias and the revolving door, respectively, influence credit ratings. Adding to this research, our study explores the role of partisan perception as a determinant of credit ratings that is not related to economic fundamentals. Our paper also relates more broadly to the literature on the determinants and consequences of credit ratings (see, for example, Becker and Milbourn (2011), Kisgen and Strahan (2010), Xia (2014), Griffin and Tang (2012), Cornaggia, Cornaggia, and Israelsen (2017), Cunha, Ferreira, and Silva (2017)).

Furthermore, our study contributes to the literature that studies how political affiliation correlates with the behavior of financial analysts, sell-side equity analysts, corporate managers, investment managers, and investors. Prior studies document that mutual fund managers who make campaign donations to the Democratic party hold less of their portfolios in companies that are deemed socially irresponsible (Hong and Kostovetsky (2012)), left-wing voters are less likely to invest in stocks (Kaustia and Torstila (2011)), sell-side equity analysts who make political contributions to the Republican Party are less likely to issue bold recommendations (Jiang, Kumar, and Law (2016)), and Republican firm managers maintain more conservative corporate policies (Hutton, Jiang, and Kumar (2014)). These studies focus on the time-invariant attributes that characterize Democrats versus Republicans, whereas we focus on how the behavior of analysts changes depending on whether their preferred party is in power. We can therefore separate the effect of partisan perception from unobserved time-invariant characteristics of individuals with different political affiliations.

Finally, our findings relate to the broader literature on belief heterogeneity, which argues that agents do not interpret public information identically and investigates the implications for asset prices (e.g., Harris and Raviv (1993), Kandel and Pearson (1995), Bamber, Barron, and Stober (1999), Banerjee and Kremer (2010), Banerjee (2011), Meeuwis et al. (2018)). Our setting allows us to provide direct evidence that Democratic and Republican analysts have

partisan models of the world and that they update their beliefs about credit risk heterogeneously in response to a given public event. Although we cannot distinguish between all possible theories of belief formation, we can exclude several stories based on our findings. For example, since we study presidential elections, which are very salient public signals, our results are unlikely to be explained by limited attention (e.g., DellaVigna and Pollet (2009), Hirshleifer, Lim, and Teoh (2009)). Moreover, analyst beliefs do not converge over the course of a presidency, which is consistent with frictions in learning (Xiong (2013)).

II. Data and Sample Construction

A. Data

The main data set used in the analysis is constructed by combining credit ratings on corporate debt issuers, press releases with analyst information, and voter registration records. We also complement the data with a variety of other data sources. The data sets are described below; further details can be found in Section I of the [Internet Appendix](#).⁵

A.1. Corporate Credit Ratings

We collect rating actions on U.S. corporate debt issuers from all three major ratings agencies: Fitch, Moody's, and S&P. Specifically, we obtain this information for S&P from S&P RatingXpress, for Moody's from Moody's Default and Recovery Database (DRD), and for Fitch from Mergent FISD.⁶ The sample period spans the first quarter of 2000 to the first quarter of 2018. We restrict the sample to post-2000 because press releases with analyst information are sparse prior to 2000. We transform credit ratings into a cardinal scale, starting with 1 for AAA (Aaa) and ending with 21 for D (C), as in Fracassi, Petry, and Tate (2016). We match each rating action (i.e., new rating, downgrade, upgrade, affirmation, internal review, reinstatement, and withdrawal) to a press release that contains the name(s) of the analyst(s) covering the firm. The press releases are collected from Moody's and Fitch's websites and from S&P's Global Credit Portal and usually contain two names, the name of the lead analyst and the name of a second analyst (often the rating committee chair or the backup analyst).

⁵ The [Internet Appendix](#) may be found in the online version of this article.

⁶ Because Mergent provides bond ratings rather than issuer ratings, we follow the procedure of Fracassi, Petry, and Tate (2016) and select a representative issuer rating after excluding bonds that are exchangeable, puttable, convertible, pay-in-kind, subordinated, secured, or guaranteed, as well as zero-coupon bonds and bonds with variable coupons.

A.2. Political Affiliation

Our political affiliation measure comes from the voter registration records from the State of Illinois, the State of New Jersey, and New York City.⁷ The voter registration records contain identifying information such as voter names, date of birth, and mailing address, the voter's party affiliation at the time of a given election, and an indicator for the election(s) in which the individual has voted. The elections covered are general, primary, and municipal elections over the period 1983 to 2017 for New York City, 1976 to 2017 for Illinois, and 2007 to 2017 for New Jersey. In Section I of the [Internet Appendix](#), we describe the information available in the voter registration records of each location in more detail.

For the purpose of our study, the voter registration data have several advantages. First, relative to commonly used data on financial contributions to political parties, candidates, and committees, found on the Federal Election Committee (FEC) website,⁸ the voter registration data cover a larger part of the population. In fact, according to a study by Hill and Huber (2017), fewer than 10% of registered U.S. voters are federal or state donors. Although these differences in sample restriction may not be as crucial when studying the influence of the political affiliation of high-profile individuals such as CEOs and board members, they are important when looking at employees not at the highest level of the firm, such as credit analysts, who are less likely to contribute financially to political campaigns. Second, while CEOs have been shown to influence the political contributions of their employees (Babenko, Fedaseyev, and Zhang (2019)), political affiliation inferred from voter registration records is less likely to be subject to such influence. Voter registration records are therefore able to isolate political beliefs from political influence and social pressure. Third, prior work shows that party registration is a good predictor of self-reported party identification. Igielnik et al. (2018) match commercial voter files, which are based on data from voter registration records, with a large-scale survey on political attitudes and voter behavior and show that the party affiliation in the commercial voter file correctly predicts the self-reported party identification for more than two-thirds of the panelists. The accuracy rate is even higher for states with party registration, such as New York.

A.3. Additional Data Sources

We rely on a variety of complementary data sources. First, to capture political polarization in the general public's views of economic conditions, we use the Daily Survey by Gallup, Inc. The Gallup Daily data cover around 1,000 individuals every day over the period 2008 to 2017. We require two main variables from the survey to capture the degree of polarization in economic views: (i) a measure of an individual's view of current economic conditions and

⁷ We use data from New York City as opposed to the State of New York, because the State of New York does not provide voter histories.

⁸ <https://www.fec.gov/>.

(ii) a measure of her political affiliation. To capture views on current economic conditions, the Gallup survey asks the following question: “How would you rate economic conditions in this country today — as excellent, good, only fair, or poor?” Responses to this question are converted into a scale that ranges from 1 (poor) to 4 (excellent). With respect to political affiliation, the Gallup survey contains two questions that allow survey respondents to be classified into Democrats, Republicans, or Independents. Our measure of political polarization in economic views is the absolute difference between the average economic views of Democrats and Republicans in a given calendar quarter.

In robustness tests, we use an alternative measure of the views of economic conditions that is based on the Michigan Survey. Specifically, we use the Current Economic Conditions Index. We provide more details on the two surveys in the [Internet Appendix](#), and in [Internet Appendix Figure IA.1](#) we plot the time series of both measures of polarization in economic views.

Next, we obtain quarterly firm-level financial information from Compustat and bond transactions from TRACE and merge them with the Mergent FISD issue file using CUSIPs. Following Fracassi, Petry, and Tate (2016), we remove bonds with special features, that is, bonds that are exchangeable, puttable, convertible, pay-in-kind, subordinated, secured, guaranteed, or denominated in foreign currency. We further drop zero-coupon and variable-coupon bonds, as well as bonds with missing maturity dates. We compute daily bond prices as the trade size-weighted average transaction price in TRACE, after applying several filters to reduce the influence of erroneous entries.⁹ We then use daily bond prices to calculate the yield to maturity and the duration of each bond. We remove bonds with a duration of less than one year. The benchmark rate that we use to compute credit spreads is based on interpolation of the yields of the two government bonds with the next-lower and the next-higher duration relative to the corporate bond.

Finally, we further supplement the data with hand-collected biographical information from online searches. We also use analysts’ first and last names to obtain additional characteristics. For example, we infer analysts’ ethnicity from their first and last names, using the Application Programming Interface (API) [name-prism.com](#) (see Ye et al. (2017) for details), and we infer the gender of analysts from their first name, using the publicly available API [genderize.io](#), as well as manual online searches.¹⁰

B. Sample Construction

We convert rating actions into an analyst-firm-quarter panel by using the most recent rating at the end of a given quarter and the analyst information

⁹ As in Dick-Nielsen (2009), we delete duplicates, reversals, cancellations, and same-day corrections. We further follow Bongaerts, Cremers, and Goetzmann (2012) and remove trades that include a commission, have a settlement period of more than five days, or a negative reported yield.

¹⁰ The API uses a large data set of first names and known genders gathered from user profiles across major social networks to predict gender. See <http://api.genderize.io/>.

from the most recent press release for the firm. To minimize measurement error in analyst assignment, we do not use analyst information from press releases older than three years as of quarter-end, and we do not assign analysts to quarters beyond the date of the final report for a given agency-firm pair. After restricting attention to analysts who work in Chicago and New York City, our sample consists of 2,402 issuers rated by 1,211 analysts.¹¹ Because we require information on political party affiliation, we further restrict the sample to analysts who can be matched to a voter registration record. We match analysts to voters as follows. In a first step, we merge analysts to voters using first name, middle initial, and last name, keeping only exact matches. In the case of duplicate matches, we try to determine the correct match based on voter age and ZIP code.¹² In a second step, we merge the remaining unmatched analysts to voter records using only their first and last names. The merging procedure is described in more detail in the [Internet Appendix](#). Our final sample contains 557 analysts, covering 1,984 firms.

To put the resulting match rate of approximately 46% ($= 557/1,211$) in context, consider the following statistics. As of November 2016, the share of registered voters among the total voting-age population of individuals aged 25 to 64 years with a bachelor's degree or higher is approximately 75.6%.¹³ We lose analysts who work in New York City but reside in Connecticut or in other parts of the State of New York. According to New York City commuter data, approximately 12% of analysts should fall into this category.¹⁴ Moreover, we lose analysts who did not update their voter registration to the state of their work location, whose names are spelled differently in press releases than in the voter records, and who match to multiple voters among whom we cannot determine a single correct match.¹⁵ Of course, these statistics have to be treated with caution, because we do not know how the population of credit rating analysts compares to the U.S. population. We nevertheless find them useful because they suggest that a match rate with voter registration records of 46% is not unreasonable.

While our analysis does not require a random sample, we would still like to understand the potential differences between our sample and the overall population of analysts and firms. First, we investigate whether the analysts that we are able to match to voter records rate different types of companies. The results, reported in Table [IA.III](#) in the [Internet Appendix](#), show that

¹¹ When the press release does not provide office information, we assume that the analyst is based in New York. Given that more than 85% of all analysts with nonmissing office location are based in New York, we believe this assumption is reasonable.

¹² Analyst age is approximated by assuming that the analyst was 21 years old upon graduation from college, following Chevalier and Ellison (1999). Information on analysts' graduation year is obtained via manual online searches.

¹³ U.S. Census Bureau. Data available at <https://www.census.gov/data/tables/time-series/demo/voting-and-registration/p20-580.html>.

¹⁴ NYC Department of City Planning. Data available at https://www1.nyc.gov/assets/planning/download/pdf/data-maps/nyc-population/acs/ctpp_p6_nyc_boro_06_10.pdf.

¹⁵ For 65 analysts, we are unable to determine a unique match out of multiple potential voter matches.

firms rated by analysts for whom we are able to obtain party affiliation are not statistically different on observable characteristics from firms rated by analysts without a matching voter record. Second, in terms of selection based on observable analyst characteristics, we do not expect analysts who are registered voters to be representative of the overall analyst population. Given that the focus of our study is to estimate the importance of political alignment with the president on the decisions of financial experts, restricting the sample to analysts who are registered voters, even if they differ from the general population of analysts, is justified. We provide a comparison of partisan analysts relative to unregistered analysts, as well as a comparison of Democratic and Republican analysts, in [Internet Appendix Table IA.IV](#).

Our analysis relies on comparisons of rating actions across agencies. We note that we do not need split-ratings, but rather split-rating actions (i.e., one rating agency changing the rating differently than another agency). Our sample contains 9,464 firm-quarters in which a firm is rated by more than one rating agency. Of these 9,464 firm-quarters, 1,617 have a split-rating action.

C. Summary Statistics

Table I and Figure 1 report summary statistics. We find that 26% of the analysts who are registered voters are unaffiliated, 39% are registered Democrats, 34% are Republicans, and 1% are registered with a party other than the Democratic or Republican party.

Figure 1, Panel A, reports the average party affiliation by Global Industry Classification Standard (GICS) sector. The sectors with the highest share of Democratic analysts are utilities (63%) and consumer staples (50%), and the sector with the highest share of Republican analysts is energy (59%). These patterns could be driven by factors such as the geography of where analysts grew up, which may influence both their party affiliation and the sectors they choose to cover. For example, analysts who grew up in New York (a “blue state”) likely had more exposure to the financial sector, whereas analysts from Texas (a “red state”) may be more familiar with energy companies. Figure 1, Panel B, shows that the percentage of Democratic analysts is higher in New York City (44%) than in Chicago (27%). Figure 1, Panel C, compares the political affiliation of analysts in the three different rating agencies. At S&P, Fitch, and Moody’s, 54%, 34%, and 33% of analysts are Democrats, respectively. The median analyst is in the sample for approximately five years.

Our main dependent variable is the quarterly change in credit rating (measured in notches). To limit the influence of outliers, we winsorize the quarterly rating change at the top and bottom 1%.¹⁶ Because credit ratings are transformed into a cardinal scale, starting with 1 for AAA (Aaa) and ending with 21 for D (C), as in Fracassi, Petry, and Tate (2016), a positive rating change

¹⁶ Table IA.IX shows that our results are robust to using unwinsorized changes (Panel A), removing firms that are in default, for which ratings are less likely to be subjective (Panel B), and to dropping rather than winsorizing outliers (Panel C).

Table I
Summary Statistics

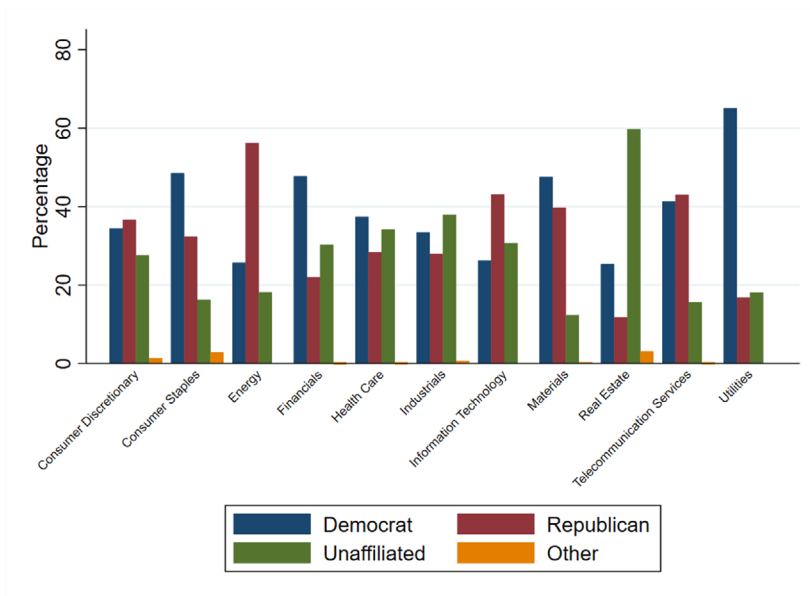
This table presents summary statistics for our key variables. The sample consists of all rating changes for U.S. corporate bond issuers by Fitch, Moody's, and S&P between 2000Q1 and 2018Q1, with available information on the analyst's political party affiliation. Observations are at the analyst-firm-quarter level. All variables are defined in the [Appendix](#).

	<i>N</i>	Mean	St.Dev.	0.25	Median	0.75
<i>Key Dependent Variables</i>						
Rating change	72,732	0.026	0.387	0.000	0.000	0.000
CAR(−1,+1) around rating changes	71,631	−6.354	158.734	0.000	0.000	0.000
CAR(−3,+3) around rating changes	71,631	−10.622	234.623	0.000	0.000	0.000
Δ Yield around rating changes	68,171	0.243	12.494	0.000	0.000	0.000
Δ Yield spread around rating changes	68,171	0.238	11.208	0.000	0.000	0.000
Δ Yield spread 4Q	33,210	−0.118	2.100	−0.571	−0.092	0.356
Δ Yield spread 8Q	28,374	−0.183	2.461	−0.750	−0.109	0.422
Δ Yield spread 12Q	23,857	−0.181	2.509	−0.856	−0.130	0.444
Accuracy 4Q	63,087	−0.406	0.907	−1.000	0.000	0.000
Accuracy 8Q	52,880	−0.714	1.180	−1.000	0.000	0.000
Accuracy 12Q	43,907	−0.947	1.318	−1.000	−1.000	0.000
<i>Key Independent Variables</i>						
Ideological mismatch	76,969	0.370	0.483	0.000	0.000	1.000
<i>Control Variables</i>						
Tenure	76,969	3.234	0.761	2.833	3.466	3.807
No. of firms covered	76,969	2.418	0.978	1.792	2.398	2.996
Votes in midterm or primary	76,969	0.754	0.431	1.000	1.000	1.000
Avg. election gap	74,678	1.643	0.558	1.253	1.580	1.962
Frequent voter	76,969	0.782	0.413	1.000	1.000	1.000
Age	76,969	3.852	0.204	3.738	3.892	4.007
Size	65,705	8.820	1.607	7.707	8.765	9.826
Investment	60,528	0.113	0.104	0.042	0.083	0.149
Leverage	65,677	0.330	0.231	0.179	0.291	0.431
Tobin's <i>Q</i>	53,993	1.530	0.683	1.065	1.316	1.764
Cash	65,692	0.084	0.095	0.017	0.052	0.117
Cash flow	59,878	0.017	0.025	0.009	0.018	0.029
Median lagged rating	74,805	11.017	3.865	8.000	10.000	14.000
Beta	49,626	1.179	0.669	0.709	1.093	1.537
Sector avg. beta	76,956	1.157	0.299	0.911	1.228	1.397
Econ. polarization Gallup	62,015	0.001	0.986	−0.866	0.114	0.615

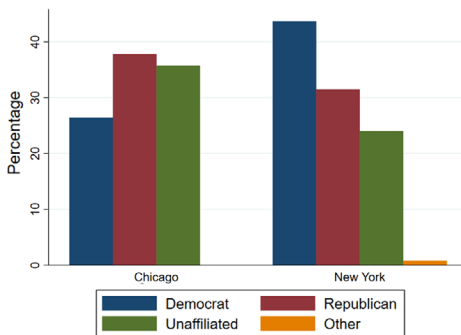
indicates a downgrade. The average credit rating change in our sample is 0.026 notches, confirming evidence from prior studies that downgrades are more common than upgrades (Dichev and Piotroski (2001), Hand, Holthausen, and Leftwich (1992), Holthausen and Leftwich (1986)). Also consistent with prior literature, 10% of the observations in our sample have a rating change (e.g., Becker and Ivashina (2014)).

Figure 2 presents the average adjusted rating change for Democratic, Republican, and unaffiliated analysts during the five presidential terms in our sample period. Adjusted rating changes are computed by taking the quarterly

Panel A. Party Affiliation by GICS Sector



Panel B. Party Affiliation by City



Panel C. Party Affiliation by Rating Agency

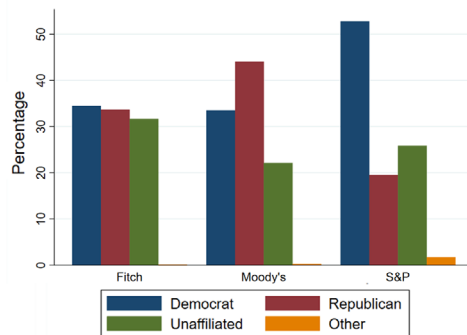


Figure 1. Party affiliation summary. The figure plots the distribution of Democratic, Republican, unaffiliated, and other analysts by GICS sector (Panel A), city (Panel B), and rating agency (Panel C).

rating change and subtracting the average rating change within the same firm-quarter across all agencies rating the firm. This measure allows us to control for the possibility that the party affiliation of the analysts covering a firm may correlate with the firm's fundamentals and investment opportunities under different administrations. Even in this univariate comparison, we observe a pattern that is consistent with our main multivariate analysis: During a Republican presidency, Republican analysts upward-adjust ratings more

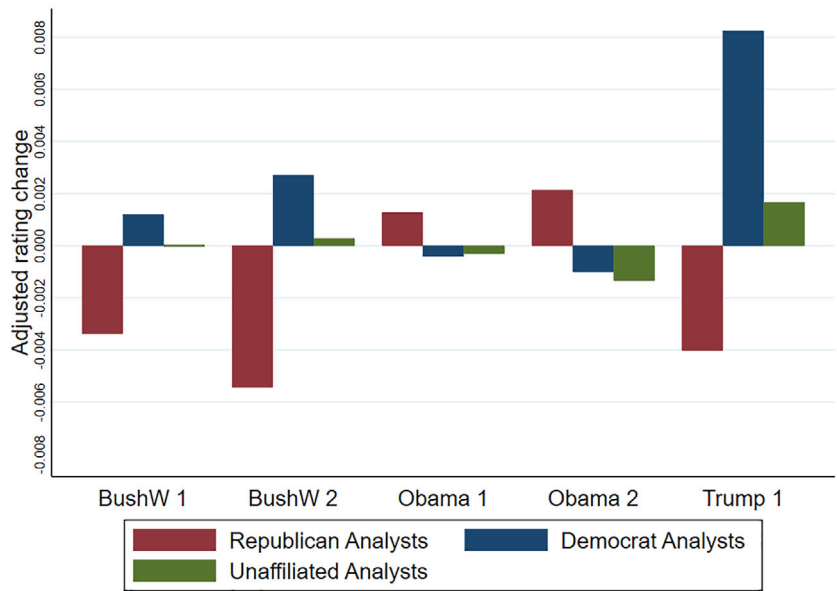


Figure 2. Average rating changes by analyst party affiliation and presidency. The figure plots the average adjusted rating change separately for Democratic, Republican, and unaffiliated analysts under each presidency during our sample period. Adjusted rating changes are computed by taking the quarterly rating change and subtracting the average rating change within the same firm and quarter.

relative to Democratic analysts. Under Obama’s presidency, the sign of this difference reverses: Republican analysts downward-adjust more than Democratic analysts. The Trump presidency is particularly interesting because the outcome of the 2016 election was unexpected, it led to the Republican party controlling both the Senate and the White House, and the two candidates involved in the election had very different views on economic policy. The disagreement between Democratic and Republican analysts is particularly large during Trump’s presidency, with Democratic analysts downward-adjusting more than Republicans. Importantly, the rating behavior of unaffiliated analysts is relatively similar under Democratic and Republican presidents. An additional pattern that emerges from Figure 2 is that the disagreement between Republican and Democratic analysts is greater during Republican administrations. This finding is consistent with existing survey evidence on households, where the partisan divide in terms of views of the economy is also stronger under Republican presidents (e.g., Pew Research Center (2019)).

III. Empirical Strategy

Measuring the influence of political alignment with the president on rating decisions by credit analysts is empirically challenging. If analysts were

randomly assigned to firms and agencies and party affiliation was randomly assigned to analysts, we could measure the effect of partisan perception by comparing the rating actions of analysts who are aligned with the president's party with the rating actions of analysts who are not aligned with the president's party. However, analysts are unlikely to be randomly assigned to firms. Analysts with a certain political ideology could specialize in sectors or firms whose fundamentals are directly affected by presidential policies (see Figure 1, Panel A, for the distribution of analyst party affiliation across sectors). For example, Republican analysts could be more likely to rate firms whose value increases under the policies of Republican presidents and decreases under the policies of Democratic presidents (e.g., oil and gas companies), and therefore downgrade more often under Democratic than under Republican administrations. In the presence of such nonrandom matching, the estimated average difference in rating actions between analysts with different party affiliations may not reflect the effect of partisan perception but rather differences in the fundamentals of the firms they cover. Second, party affiliation is not randomly assigned to analysts and may be correlated with other time-invariant analyst characteristics, such as upbringing, education, prior work experience, or attitudes toward certain industries or firms. Third, credit analysts may not be randomly assigned to rating agencies. As Figure 1, Panel C, shows, the mix of Democratic and Republican analysts varies substantially across agencies. If political cycles correlate with asset returns (see Pástor and Veronesi (2018)), and rating agencies' methodologies differ in how they incorporate economic variables into their models, differences in ratings between agencies over the political cycles might not be due to partisan perception but to the nonrandom composition of political affiliations across agencies.

Our identification strategy addresses the confounding factors above by regressing the rating change for firm f rated by analyst i in quarter t on firm $\times$ quarter fixed effects, political affiliation fixed effects, as well as agency $\times$ quarter fixed effects,

$$\Delta R_{ift} = \alpha_{ft} + \alpha_{jt} + \alpha_p + \beta \text{Ideological mismatch}_{it} + \gamma' X_{it} + \epsilon_{ift}, \quad (1)$$

where ΔR_{ift} is the quarterly rating change, j denotes the rating agency, p denotes the analyst's political affiliation (Democrat, Republican, Unaffiliated, and Other), and $\text{Ideological mismatch}_{it}$ is an indicator equal to one if the analyst's party affiliation does not match the party of the elected president in quarter t . In a presidential election quarter, we define *ideological mismatch* using the newly elected president, because the election result is known by the end of the quarter. The vector X_{it} includes analyst-level controls, including analyst tenure and the number of firms rated by the analyst.

In our baseline definition, *ideological mismatch* is equal to one for analysts whose party affiliation does not match the president's, and zero for analysts whose party affiliation matches that of the president and for unaffiliated analysts. This definition results in 37% of analyst-firm-quarters having misalignment with the president's party (see Table I). Due to the inclusion of

party affiliation fixed effects in all regressions, the coefficient on *ideological mismatch* will be identified based only on Republican and Democratic analysts, because their party switches at least once from aligned to misaligned with the president.¹⁷ The main benefit of including unaffiliated analysts in our analysis is that doing so allows us to estimate the fixed effects and coefficients on control variables more precisely. In [Internet Appendix Table IA.V](#), we obtain very similar results if we only keep Democratic and Republican analysts.

We study rating changes instead of rating levels for two main reasons. First, rating changes allow us to better isolate the decisions of the current analyst from other confounding factors, such as the influence of the previous analyst. Second, rating levels are sticky because analysts do not adjust them frequently. In [Section C.1](#) below we show that in levels, the effect is there but takes some time to show up, consistent with slow adjustment of ratings.

By including firm $\times$ quarter fixed effects, we compare analysts who cover the same firm at the same point in time. And this ensures that our results are not driven by analysts with ideological mismatch covering different types of firms. Including agency $\times$ quarter fixed effects addresses concerns that different methodologies of credit agencies might be correlated with the political cycle during our sample period. We also present specifications including agency $\times$ sector instead of agency $\times$ quarter fixed effects to address any concerns related to agencies with different compositions of Democratic and Republican analysts having different rating models for different industries. Finally, our estimator of partisan perception is, by construction, orthogonal to the baseline effect of the analyst's political affiliation and hence to a number of unobserved time-invariant analyst characteristics that may be correlated with party affiliation. Thus, our estimator of partisan perception captures how the behavior of Democratic and Republican analysts changes depending on whether their preferred party occupies the White House, and not time-invariant characteristics correlated with party affiliation.

Alternatively, it would be possible to control for unobserved analyst heterogeneity via analyst fixed effects. However, for unobserved time-invariant analyst characteristics to be a concern, they would have to be correlated with our measure of *ideological mismatch* and, at the same time, be orthogonal to party affiliation. It is difficult to think of analyst characteristics that would meet these two criteria. Moreover, including analyst fixed effects imposes costs, especially in tests in which we partition the data further, because the tenure of 41% of analysts does not span both a Democratic and a Republican president and these analysts do not experience any within-analyst change in misalignment. Despite these limitations, we report all main specifications using analyst fixed effects in the [Internet Appendix](#) and discuss these results in [Section C](#) below. In [Section C](#) we also repeat our main analysis after including additional analyst characteristics as well as their interaction with

¹⁷ Because unaffiliated analysts and analysts with other party affiliations never change from aligned to misaligned with the president, whether we code them as aligned or misaligned does not affect our estimate of β in equation (1).

an indicator for Democratic presidents. We include characteristics that are known to be important predictors of political affiliation, namely, ethnicity, gender, and age. Across all three specifications, the coefficient estimate on *Ideological mismatch* is remarkably stable.

We double-cluster standard errors at the analyst-firm level throughout the analysis. In Section C, we show that our results are robust to triple clustering by firm, analyst, and quarter.

A. Can Individual Analysts Influence Ratings?

A necessary condition for analysts' political ideology to affect ratings is that the ratings process needs to provide sufficient room for individual analysts to affect the rating of an issuer. Whether individual analysts can affect ratings is not obvious given that the final rating decision is taken by a committee. Upon receiving a rating application from a potential customer, rating agencies typically assign a lead analyst to the ratings process. The lead analyst meets with the customer to discuss relevant information, which she subsequently analyzes with the help of an analytical team. She then proposes a rating and provides a rationale to the rating committee, which consists of a number of credit risk professionals determined by the analyst in conjunction with the committee chair. Once the rating committee has reached its decision, the rating agency communicates the outcome to the customer and publishes a press release.¹⁸ The ratings process therefore provides ample opportunities for individual analysts to influence the final rating, even if the final decision is taken by a committee. Lead analysts guide meetings with the customer, request and interpret information, and play a key role in the rating committee by proposing and defending a rating recommendation based on their own analysis. In addition, the rating committee chair serves a special role by influencing the composition of the rating committee and acting as committee moderator.

The extent to which individual analysts are able to influence ratings is ultimately an empirical question. Fracassi, Petry, and Tate (2016) attribute a substantial part of the variation in corporate bond ratings to individual analysts. In particular, they show that individual analysts explain 30% of the within-firm variation in ratings. For securitized finance ratings, Griffin and Tang (2012) provide evidence that collateralized debt obligations ratings by a major credit rating agency frequently deviated from the agency's main model, reflecting room for subjectivity in the ratings process. In addition, Kisgen et al. (2020) and Kempf (2020) document that the ratings decisions of lead analysts and committee chairs predict both internal promotions and external hiring by

¹⁸ See Fracassi, Petry, and Tate (2016) and https://www.moody.com/sites/products/ProductAttachments/mis_ratings_process.pdf for a description of the ratings process at Moody's, https://www.standardandpoors.com/en_AU/delegate/getPDF?articleId=2053416&type=COMMENTS&subType=REGULATORY for the ratings process at S&P, and <https://www.fitchratings.com/site/dam/jcr:b05b5bd2-0443-4338-815b-81a809840e65/Form%2025-101F1%20Item%205.pdf> for the ratings process at Fitch.

investment banks, suggesting that, by revealed preferences, these parties see valuable information in analysts' rating decisions.

B. Why Do Corporate Credit Analysts Provide a Useful Setting?

Corporate credit rating analysts provide a suitable setting to study the effect of partisan perception on economic behavior in a high-stakes environment for a number of reasons. First, this setting is one of the few in which decisions can be linked to the identity of the individual decision maker. Second, credit ratings have been shown to affect firms' cost of financing (Fracassi, Petry, and Tate (2016)), as well as their financial policy and investment decisions (Chernenko and Sunderam (2011), Almeida et al. (2017)) and thus the views and actions of credit rating analysts have real consequences for the rated firms. Third, because multiple analysts can rate the same firm at the same point in time, we are able to control for differences in the type of firms that analysts rate. Finally, one of the challenges in addressing our question is that we need to match individual analysts to voter registration data. The geographical location of credit analysts is very concentrated in New York City and Chicago, which reduces problems related to heterogeneous data availability and data quality in voter registration records across different U.S. states.

IV. Analysts' Partisan Perception and Credit Rating Actions

This section presents our main results. We first document that partisan perception has a significant influence on analysts' credit rating decisions by showing that analysts whose party affiliation does not match the president's are more likely to adjust ratings downward. This result is robust to various measures of rating changes, sample restrictions, and estimation methods. Next, we focus on presidential elections. We document a sharp divergence in the rating actions of Democratic and Republican analysts during the month of the 2016 presidential election, with Democratic analysts downward-adjusting ratings more than Republican analysts.

A. Main Results

We estimate the regression from equation (1) with different sets of fixed effects and report the results in Table II. Specifically, we regress quarterly ratings changes on *Ideological mismatch*, an indicator equal to one if the analyst's party does not match that of the current president and zero otherwise, and controls. We include party affiliation fixed effects in all regressions.

We begin by including agency fixed effects in addition to firm $\times$ quarter fixed effects and analyst's party affiliation fixed effects, because Figure 1 documents pronounced differences in political ideology across agencies. The results, reported in column (1), suggest that partisan perception affects analysts' rating behavior. Specifically, analysts who do not support the president's party are more likely to adjust ratings downward by 0.0168 notches relative to analysts

Table II
Partisan Perception and Rating Changes

This table regresses quarterly rating changes on *Ideological mismatch*, an indicator equal to one for analysts whose party affiliation does not match the president's party and zero otherwise. All variables are defined in the [Appendix](#). *t*-statistics, reported in parentheses, are based on standard errors that allow for double-clustering at the firm-analyst level.

	Rating Change		
	(1)	(2)	(3)
Ideological mismatch	0.0168 (4.18)	0.0170 (4.26)	0.0134 (3.79)
Tenure	0.0001 (0.03)	-0.0006 (-0.23)	-0.0004 (-0.15)
No. of firms covered	-0.0001 (-0.10)	0.0001 (0.05)	-0.0000 (-0.02)
Observations	49,792	49,792	49,790
R^2	0.804	0.804	0.808
Firm $\times$ Quarter FE	Yes	Yes	Yes
Agency FE	Yes	No	No
Agency $\times$ Sector FE	No	Yes	No
Agency $\times$ Quarter FE	No	No	Yes
Party affiliation FE	Yes	Yes	Yes

who are aligned with the president's party. The economic and statistical significance of the effect of partisan perception remain high as we tighten identification to include agency $\times$ sector (column (2)) and agency $\times$ quarter fixed effects (column (3)). The estimate in column (3) suggests that an analyst who is misaligned with the president's party on average downward-adjusts ratings more by 0.0134 notches. Relative to the average absolute rating change of 0.118 notches, this change is an economically sizable increase of 11.4%. Over a four-year presidency (i.e., 16 quarters), these estimates imply that analysts who are misaligned with the president downward-adjust the rating of the average firm by 0.21 ($= 0.0134 \times 16$) notches more than aligned analysts.¹⁹ As a reference point, a one-notch downgrade would be equivalent to moving from A to A- on the credit rating scale.

Comparing the effect of the analyst's political alignment with the president to other unobserved factors at the analyst level is also informative. We measure the importance of unobserved analyst factors using the distribution of analyst fixed effects (i.e., we estimate the regression reported in Table II, column (3), with analyst fixed effects). We find that the effect of *ideological mismatch* is roughly equivalent to moving from the 50th to the 75th percentile of the analyst fixed-effects distribution.

¹⁹ Since coefficient β in equation (1) captures the difference between misaligned and aligned analysts during the *average* quarter, we need to multiply by 16 quarters to obtain the cumulative effect over a presidential term.

The effect of partisan perception that we document is comparable to other nonfundamental factors influencing information production at rating agencies identified in the literature. For example, the effect of competition among rating agencies documented by Becker and Milbourn (2011) is equal to 0.19 notches, the revolving-door effect documented by Cornaggia, Cornaggia, and Xia (2016) is equal to 0.23 notches, and the home-bias effect documented by Cornaggia, Cornaggia, and Israelsen (2019) is equal to 0.10 notches on average.²⁰

Our set of high-dimensional fixed effects eliminates a lot of potentially confounding variation and allows us to overcome some of the central challenges in empirical studies of partisan behavior. Most importantly, firm $\times$ quarter fixed effects address the possibility that the relationship between partisan perception and rating actions may be confounded by nonrandom matching of analysts to firms. Second, by including agency $\times$ quarter fixed effects, we can remove differences in rating methodologies across rating agencies. Third, by including party affiliation fixed effects, we remove any time-invariant differences across individuals with different party affiliations and hence can isolate changes in their rating behavior as the president changes.

In Table IA.V, we repeat our main analysis but report the effects separately for Democratic and Republican analysts (Panel A) as well as for Democratic, Republican, and unaffiliated analysts (Panel B). In Panel A, the coefficient on *Democrat* which captures the difference in the average rating adjustment between Democratic and Republican analysts under Republican presidents, is equal to 0.0268 notches (column (3)). Under Democratic presidents, the difference between Democratic and Republican analysts is equal to -0.0031 notches ($= 0.0268 - 0.0299$) and statistically insignificant. Hence, consistent with Figure 2, we find that the difference between Democratic and Republican analysts is more pronounced under Republican presidents than under Democratic presidents. Moreover, Panel B further shows that Democratic analysts behave more similarly to unaffiliated analysts, as the difference between Democratic and unaffiliated analysts is almost never statistically significant.

A potential concern could be that our *ideological mismatch* variable picks up the effect of other analyst characteristics that may be correlated with party affiliation. Such unobservable characteristics would pose a threat to our identification only if they can explain differential behavior under Democratic versus Republican administrations. What characteristics that might be is not obvious. Nonetheless, to address this potential threat directly, Internet Appendix Table IA.VI repeats the analysis from Table II while including additional analyst characteristics as well as their interaction with an indicator for Democratic presidents. We include characteristics that are known to be important predictors of political affiliation, namely, ethnicity, gender, and age. Across all three specifications, the coefficient estimate on *Ideological mismatch* is remarkably stable.

²⁰ This comparison has to be taken with a grain of salt, because the various studies differ in their econometric specification and empirical setting.

B. Presidential Elections

B.1. All Presidential Elections

To further support the interpretation that our results reflect the effect of analysts' partisan perception, we conduct an event study around presidential elections. We focus on presidential elections in which the party of the president changes (i.e., the 2000, 2008, and 2016 elections) to isolate the effect of a change in ideological alignment on rating changes. Specifically, we estimate the regression,

$$\Delta R_{ift} = \alpha_{ft} + \alpha_{jt} + \sum_{\tau=-4}^{\tau=+4} \beta_{\tau} \text{Mismatch}_{i,\tau=0} \times D_t^{\tau} + \epsilon_{ift}, \quad (2)$$

where $\text{Mismatch}_{i,\tau=0}$ is an indicator equal to one for analysts who switch from aligned to misaligned with the president around the election in $\tau = 0$, and D_t^{τ} denotes calendar quarter dummies. We restrict the analysis to Democratic and Republican analysts only. The parameters α_{ft} and α_{jt} correspond to firm $\times$ quarter and agency $\times$ quarter fixed effects, respectively.

Figure 3 plots the results. We find small and insignificant differences between aligned and misaligned analysts during the quarters leading up to the election. During the election quarter, we see a striking and statistically significant divergence in rating actions, with misaligned analysts downward-adjusting ratings more by 0.0601 notches. Overall, we see larger downward adjustments by misaligned analysts in the postelection period (quarters 0 to +4) than in the preelection period (quarters -4 to -1), with the difference being statistically significant at the 10% level.

B.2. 2016 Presidential Election

We next focus on the 2016 election because it is unique—the outcome was unexpected and the two candidates had very different positions on economic issues. Moreover, unlike the 2008 election, the 2016 election did not overlap with major economic events.

Figure 4 plots the difference in the average rating adjustment between Democratic and Republican analysts around the 2016 election. We estimate the same regression as in equation (2), but we use monthly rather than quarterly data to better isolate the effect of the election.

The difference in the average rating adjustment between Democrats and Republicans is close to zero prior to the election. The difference then spikes significantly during the election month, November 2016, indicating that Democratic analysts downward-adjusted firms more by 0.034 notches. Despite the data limitations related to this event study and the fact that rating changes are infrequent events, the results indicate a significant change in rating behavior between Democratic and Republican analysts around the 2016 election. We see a small reversal during February 2017, with Democratic analysts upward-adjusting ratings slightly more than Republican analysts,

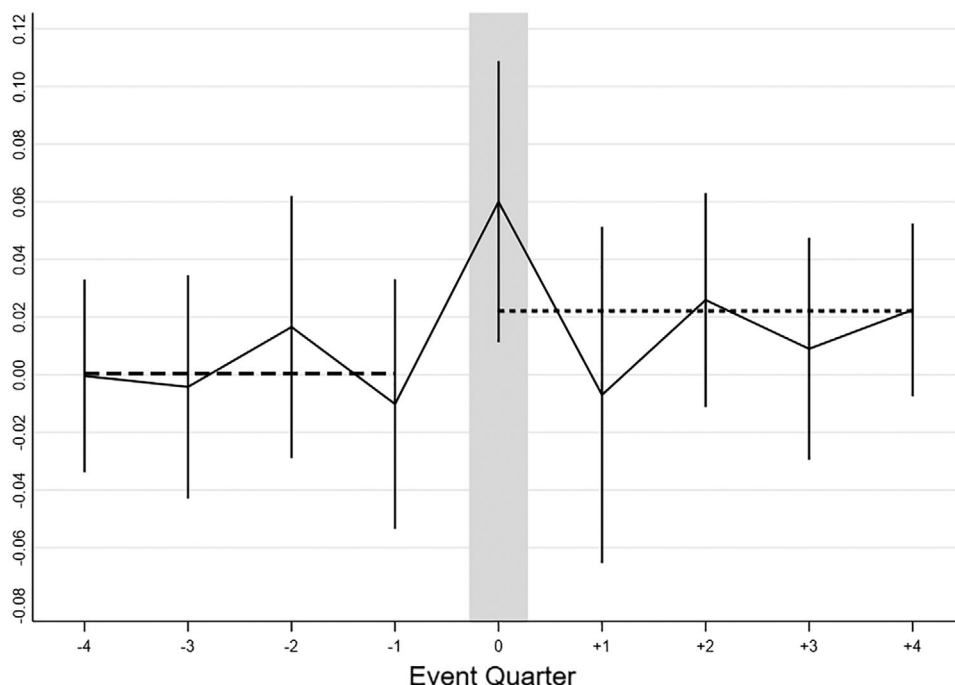


Figure 3. Quarterly event study around presidential elections. The figure plots the difference in the average quarterly rating adjustment between misaligned and aligned analysts around presidential elections. Specifically, we plot the coefficient estimates for β_t from equation (2). The corresponding 95% confidence intervals are based on standard errors that are double-clustered at the firm-analyst level. The dashed line indicates the average difference in the rating change prior to the election (i.e., -4 to -1) and the dotted line indicates the average difference post election (i.e., 0 to $+4$).

although the difference is not statistically significant. When we expand the event window to span the period 2015Q1 to 2018Q1, we do not find that this reversal continues. [Internet Appendix Table IA.XII](#) reports results from this analysis using quarterly data. We find that Democratic analysts, on average, downward-adjust credit ratings by 0.0245 notches more than Republicans in the post-Trump period relative to the pre-Trump period.

Due to the unexpected outcome of the election, the event study allows us to rule out several alternative stories. Given the evidence that economic conditions can influence voting behavior (see Lewis-Beck and Stegmaier (2000) for a review of the relevant literature), a potential concern about our main analysis could be that the political affiliation of analysts might change in response to presidential policies or economic conditions. The event study mitigates such reverse-causality concerns, given that political attitudes of analysts are unlikely to change discontinuously during the election month. It is also unlikely that we are picking up disagreement about other economic

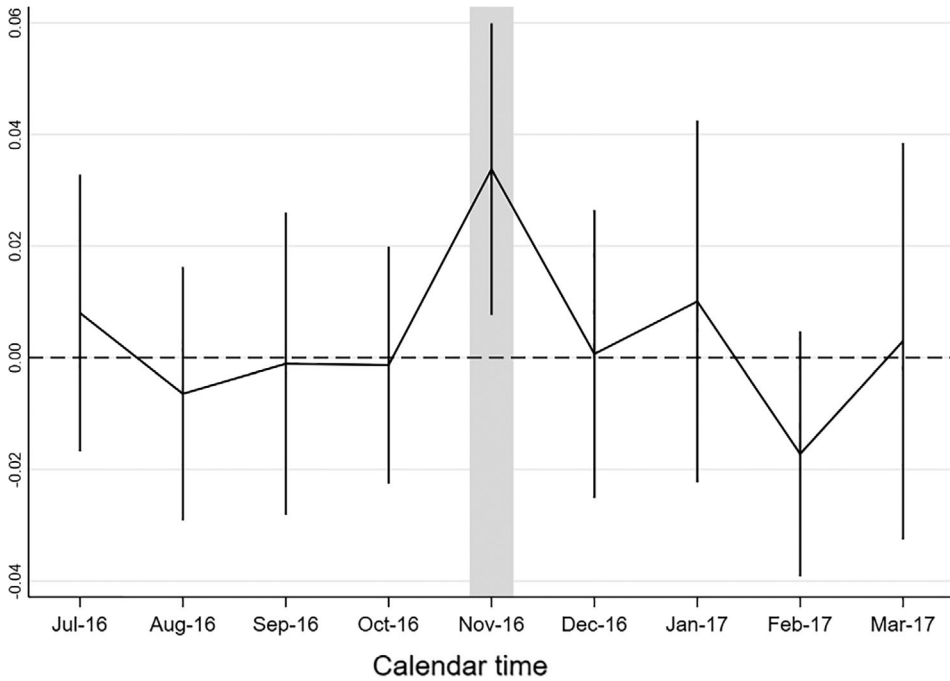


Figure 4. Monthly event study around the 2016 presidential election. The figure plots the difference in the average monthly rating adjustment between Democratic and Republican analysts around the 2016 presidential election. Specifically, we plot the coefficient estimates for β_τ from the regression

$$\Delta R_{ift} = \alpha_{fq} + \alpha_{jq} + \sum_{\tau=Jul16}^{\tau=Mar17} \beta_\tau Democrat_{it} \times D_t^\tau + \sum_{\tau=Jul16}^{\tau=Mar17} \delta_\tau \times D_t^\tau + \epsilon_{ift}, \quad (6)$$

where $Democrat_{it}$ is an indicator equal to one for analysts who are affiliated with the Democratic party and zero for Republican analysts, α_{fq} and α_{jq} are firm $\times$ quarter and agency $\times$ quarter fixed effects, respectively, and D_t^τ are calendar-time dummies. The corresponding 95% confidence intervals are based on standard errors that are double-clustered at the firm-analyst level.

events unrelated to the presidential election, because the election month did not coincide with other major events.

C. Robustness Tests

Table III presents additional robustness tests of the main result in Table II and shows our main result is robust to a battery of additional tests. Unless otherwise mentioned, we report results for the specification in Table II, column (3), and we suppress all control variables for brevity.

In Panel A, we alter the definition of the dependent variable. To mitigate the concern that our main result could be driven by outliers in the dependent variable, we modify the rating-change variable to take only three possible

Table III
Robustness

This table presents robustness tests. The baseline regression refers to specification (3) from Table II. For brevity, we only report coefficients of interest and suppress control variables. In Panel A, *Rating-change indicator* is equal to 0 if the quarterly rating change is zero, -1 if the rating change is negative (i.e., upgrade), and $+1$ if the rating change is positive (i.e., downgrade). *Downgrade (Upgrade)* is an indicator equal to one if the quarterly rating change is positive (negative) and zero otherwise. In Panel B, we use alternative definitions of *ideological mismatch*, using party affiliation from presidential elections only (first row), and adding party affiliation information from analysts' political contributions. In Panel C, *Triple clustering by analyst, firm, and quarter* refers to triple-clustering standard errors at the analyst-firm-quarter level. *Firm-agency level* refers to a regression of quarterly rating changes on average *ideological mismatch*, after collapsing the data at the firm-agency-quarter level and averaging *ideological mismatch* across all analysts rating the same firm for the same rating agency in the same quarter. *Weighted least squares* refers to a weighted-least-squares regression in which weights are proportional to the total lagged book assets of the rated firm. *Add agency $\times$ sector $\times$ quarter FE* replaces agency $\times$ quarter fixed effects with agency $\times$ sector $\times$ quarter fixed effects. *Add NBER Recession $\times$ Party affiliation FE* replaces party affiliation fixed effects with party affiliation $\times$ recession dummy fixed effects. Recessions are defined following the National Bureau of Economic Research (NBER). *t*-statistics are based on standard errors that allow for double-clustering at the firm-analyst level (except for the value in the row *Firm-agency level*, where standard errors are clustered at the firm level).

	Coeff.	<i>t</i> -statistic
Baseline	0.0134	3.79
Panel A: Alternative dependent variables		
Rating-change indicator	0.0110	3.88
Downgrade	0.0064	3.24
Upgrade	-0.0046	-2.21
Panel B: Alternative definitions of ideological mismatch		
Use only party affiliation from presidential elections	0.0144	2.75
Add party affiliation from political contributions	0.0135	4.36
Panel C: Estimation		
Triple-cluster standard errors (analyst, firm, and quarter)	0.0134	4.33
Firm-agency level	0.0301	3.81
Weighted least squares	0.0125	3.35
Add agency $\times$ Sector $\times$ Quarter FE	0.0088	2.82
Add NBER recession $\times$ Party affiliation FE	0.0150	4.03

values: $+1$ for downgrades, 0 for no change, and -1 for upgrades. The results are very similar and the statistical significance, if anything, increases. When we separate the propensity to upgrade versus downgrade, we find that the effect is coming from both sides: Analysts who are not aligned with the White House are both more likely to downgrade and less likely to upgrade.

Panel B shows results for alternative definitions of mismatch. We first use a definition of mismatch that relies only on party affiliation information from presidential elections, as opposed to all elections. We also test a definition of mismatch that complements our party affiliation from voter registration

records with party information from analysts' financial contributions to political parties, candidates, and committees.²¹ Both alternative definitions of *ideological mismatch* yield results that are similar to our baseline.

In Panel C, we assess the robustness of our results with respect to alternative estimation methods. Triple-clustering by firm, analyst, and quarter leads to smaller standard errors. We also verify that similar results are obtained if we estimate the regression in equation (1) at the agency-firm-quarter level instead of the analyst-firm-quarter level, that is, if we compute the average *ideological mismatch* across all analysts from the same rating agency covering the same firm at the same point in time. We next estimate a weighted regression in which the weights are proportional to the total book assets of the rated firm. Our results are also robust to replacing agency $\times$ quarter fixed effects with agency $\times$ sector $\times$ quarter fixed effects, to address the possibility that rating agencies with different compositions of Democratic and Republican analysts might use different rating methodologies in different sectors across Democratic or Republican presidencies. Furthermore, our results do not change when we introduce National Bureau of Economic Research (NBER) recession interacted with party affiliation fixed effects to mitigate the concern that our baseline results reflect a differential response of Democratic and Republican analysts to recessions, as opposed to a differential response to different White House administrations.

In Table IA.VII, we also show that our results are robust to including analyst fixed effects. This finding is important because it reinforces our argument that we are capturing the effect of partisan perception rather than time-invariant characteristics of the analyst that may be correlated with party affiliation. While, unsurprisingly, including analyst fixed effects tends to increase standard errors, the coefficient estimates remain relatively stable. Compared to Table II, the point estimates are 15% smaller, on average, with analyst fixed effects. In our stock price and bond yield regressions, which we discuss in Section V, the point estimates are 27% larger, on average, with analyst fixed effects. Internet Appendix Figure IA.4 shows that the quarterly event study around presidential elections looks similar if we add analyst fixed effects, with the post-pre difference significant at the 5% level.

In Table IA.XI, we also test whether *ideological mismatch* with the party that controls the U.S. Senate or the U.S. House of Representatives matters for rating decisions. In column (1), we find that analysts misaligned with the party majority in the Senate downward-adjust ratings more. We do not find a significant effect from misalignment with the party majority in the House of Representatives (column (2)). When we add misalignment with the president as an additional control in columns (3) and (4), neither *ideological mismatch* with the Senate nor *ideological mismatch* with the House of Representatives

²¹ Data on financial contributions are obtained from Stanford's Database on Ideology, Money in Politics, and Elections (DIME) (see Bonica (2016)). We are able to match 57 analysts to a federal- or state-level contribution. In the Internet Appendix, we provide additional information regarding the merging procedure.

has incremental explanatory power. This insignificance of the party in control of Congress is consistent with existing studies of political cycles and stock returns (Santa-Clara and Valkanov (2003), Pástor and Veronesi (2018)), as well as with prior studies of political cycles and GDP growth (Blinder and Watson (2016)). We do not have a complete explanation for why disagreement with the president matters, whereas disagreement with Congress does not matter incrementally. We speculate that this result could be driven by the party of the U.S. president being more salient to analysts than the identity of the party that controls Congress. According to a Gallup poll from 2014, only 41% of the surveyed registered voters were able to correctly identify the majority in both the Senate and the House (Saad (2014)). Although this percentage could be different in the population of credit rating analysts, it still highlights that the party of the president is more salient to voters than the party majority in Congress. Moreover, only 32% of registered voters prefer one-party control of Congress (Jones (2014)), indicating they do not have a strong preference for their own party controlling Congress.

Figure IA.2 plots our coefficient estimates after sequentially excluding each GICS sector that represents at least 5% of our total observations. The coefficient estimate is remarkably consistent across all specifications, suggesting that our main result is not driven by a single sector. Finally, in Figure IA.3, we find that the effect of partisan perception is fairly consistent across issuers of different credit quality.

C.1. Robustness to Using Rating Levels

We also assess the robustness of our main results to studying rating levels instead of rating changes. Figure 5 repeats the analysis of Figure 3, but uses the quarter-end rating level as the dependent variable. The figure shows a change in the difference in rating levels between aligned and misaligned analysts after the election, as is evident from the change in slope following the election. Moreover, consistent with Figure 3, we observe that the effect on the rating level is gradual. It takes a few quarters—until $t = 2$ —for the difference in rating levels to switch sign. The difference between the average difference in rating levels is statistically significant at the 10% level if we compare the average in quarters 0 to +4 to the average in quarters −4 to −1, and is statistically significant at the 5% level if we allow for a gradual response and if we exclude quarters 0 and +1.

We further study the effect of alignment with the president on rating levels over the full sample period. We use the same specification as in Table IA.V and regress the quarter-end rating on an indicator for Democratic analysts and an interaction between Democratic analysts and Democratic presidents. Table IA.VIII presents the results. Using the estimates in Panel A, column (3), we observe that, relative to Republican analysts, Democratic analysts are 0.137 notches more positive under Democratic presidents than under Republican presidents (see coefficient on the interaction term *Democrat* $\times$ *DemPresident*). The estimated effect of 0.137 notches is lower than our baseline estimate

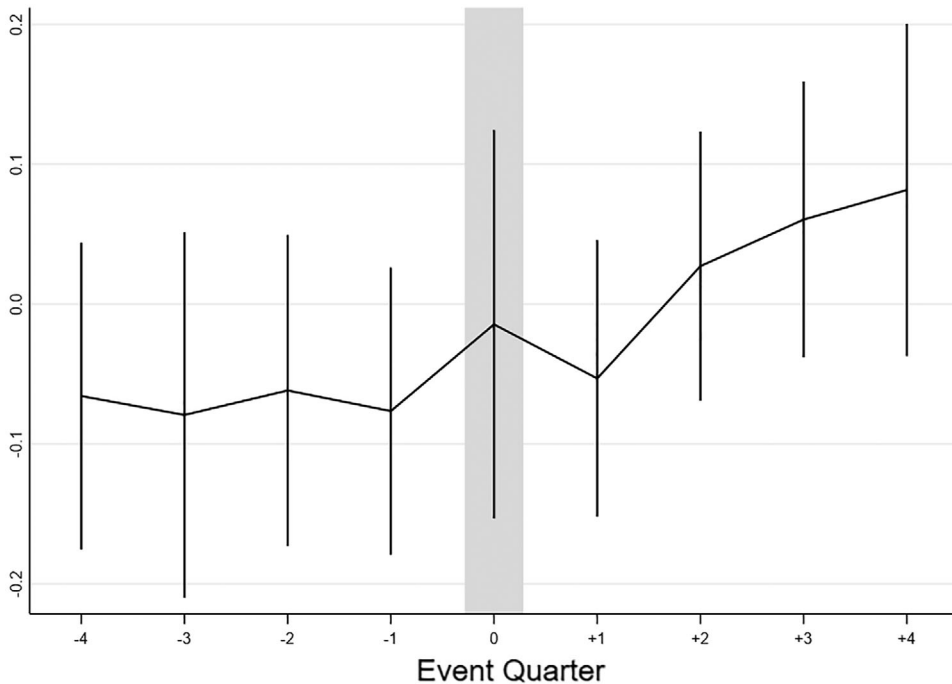


Figure 5. Quarterly event study around presidential elections: Rating levels. The figure repeats the analysis presented in Figure 3 after replacing the quarterly rating change with the quarter-end rating level as the dependent variable.

of 0.21 notches mainly due to the slow adjustment of ratings. As we show in Panel B, the coefficient gets closer to our baseline effect of 0.21 notches if we remove quarters close to the election (here we remove quarters 0 and +1). Specifically, we obtain an effect of 0.184 notches in Panel B, column (3).

In sum, these results highlight the robustness of our main results to using rating levels.

D. Heterogeneity across Analysts

We next test for heterogeneous effects across analysts. Specifically, we investigate whether the effect of political alignment with the president is more pronounced among analysts who are politically active.

In Table IV, we test three alternative proxies for political activeness based on how frequently the analyst votes. First, we identify analysts who have voted in a primary or midterm election. Second, we compute the average time gap (in quarters) between the elections in which the analyst votes. Third, we create an indicator variable *Frequent voter* that is equal to one for analysts who have voted in a midterm or primary election, or for whom the the average election gap is in the bottom quartile across all analysts within

Table IV
Interaction with Voting Frequency

This table regresses quarterly rating changes on *ideological mismatch* as well as interactions with measures of voting frequency. *Votes in midterm or primary* is an indicator equal to one for analysts who have voted during midterm or primary elections in the past and zero otherwise. *Avg. election gap* is the logarithm of the average time gap (in quarters) between elections in which the analyst has voted. *Frequent voter* is an indicator equal to one for analysts who are either in the bottom quartile of the average election gap across all analysts in a given quarter, or for whom *Votes in midterm or primary* is equal to one, and zero otherwise. *Age* is the logarithm of the age of the analyst as of the end of the quarter, and is standardized to have a mean of zero and a standard deviation of one. *t*-statistics, reported in parentheses, are based on standard errors that allow for double-clustering at the firm-analyst level.

	Rating Change		
	(1)	(2)	(3)
Ideological mismatch	0.0014 (0.21)	0.0288 (3.69)	0.0003 (0.04)
Votes in midterm or primary	0.0008 (0.25)		
Mismatch × Votes in midterm or primary	0.0140 (2.21)		
Avg. election gap		0.0003 (0.12)	
Mismatch × Avg. election gap		-0.0102 (-2.25)	
Frequent voter			-0.0013 (-0.40)
Mismatch × Frequent voter			0.0151 (2.33)
Age	-0.0001 (-0.08)	-0.0006 (-0.38)	0.0001 (0.05)
Mismatch × Age	-0.0007 (-0.25)	-0.0006 (-0.20)	-0.0009 (-0.33)
Tenure	-0.0004 (-0.15)	-0.0006 (-0.21)	-0.0004 (-0.15)
No. of firms covered	-0.0006 (-0.45)	-0.0004 (-0.28)	-0.0005 (-0.39)
Observations	49,790	47,454	49,790
R ²	0.808	0.808	0.808
Firm × Quarter FE	Yes	Yes	Yes
Agency × Quarter FE	Yes	Yes	Yes
Party affiliation FE	Yes	Yes	Yes

a given quarter, and to zero otherwise (column (3)). Because some of these measures are correlated with voter age, we also control for the interaction between voter age and *ideological mismatch*. All three measures indicate that the effect of partisan bias is substantially stronger for analysts who vote more frequently. The importance of voting frequency supports the interpretation that our results reflect differences in partisan perception and further raises the bar for alternative explanations.

V. Price and Real Effects of Analysts' Partisan Perception

Our results so far indicate that partisan perception affects corporate credit ratings. In this section we gauge the potential consequences of this result on firms' cost of financing and investment decisions. First, regarding price effects, we document that firms that are rated by misaligned analysts experience a greater reduction in their stock prices and a larger increase in their bond yields compared to firms rated by aligned analysts. Second, to explore the effect of partisan perception on firm investment, we examine the difference in the investment policies of firms rated by aligned versus misaligned analysts in the quarters around presidential elections. Our results suggest that companies rated by misaligned analysts have lower capital expenditure following the elections.

A. Stock Prices

A number of studies document the reaction of common stock prices to credit rating changes. The general conclusion of this literature, starting with Holthausen and Leftwich (1986), is that downgrades are associated with significant negative abnormal stock returns, even after eliminating observations that contain potentially contaminating concurrent news releases. Little evidence supports the assertion that upgrade announcements trigger significant abnormal returns.

We replicate these findings for our sample of rating-change announcements. Retaining all rating actions by analysts that we can match to a voter record, we compute abnormal stock returns around the date of the rating change reported in Moody's DRD, S&P RatingXpress, and Mergent, respectively. Abnormal returns are calculated using the Fama and French (1993) and Carhart (1997) model estimated over trading days $(-300, -50)$ relative to the rating change. In Figure 6, we plot the abnormal returns in the 21 days around the upgrade and downgrade announcements in our sample. We also remove any rating changes around which the firm makes an earnings announcement or an M&A announcement inside the $(-10, +10)$ window.²² Consistent with prior studies (e.g., Holthausen and Leftwich (1986), Hand, Holthausen, and Leftwich (1992)), we find large negative average abnormal returns (-1.9%) in the three days around rating downgrades, and very small abnormal returns (0.2%) around upgrades after excluding concurrent events. Also consistent with prior evidence, we find a substantial anticipation effect in stock prices prior to downgrades, indicating either information leakage or deteriorating creditworthiness prior to the downgrade.

Given our finding that analysts whose party affiliation does not match that of the president downward-adjust more frequently, a question that rises is how well does the market "correct" for analysts' partisan perception. Is the stock market reaction to downgrades muted when the downgrades are issued by

²² Earnings announcement dates are obtained from IBES and M&A announcement dates from SDC Platinum.

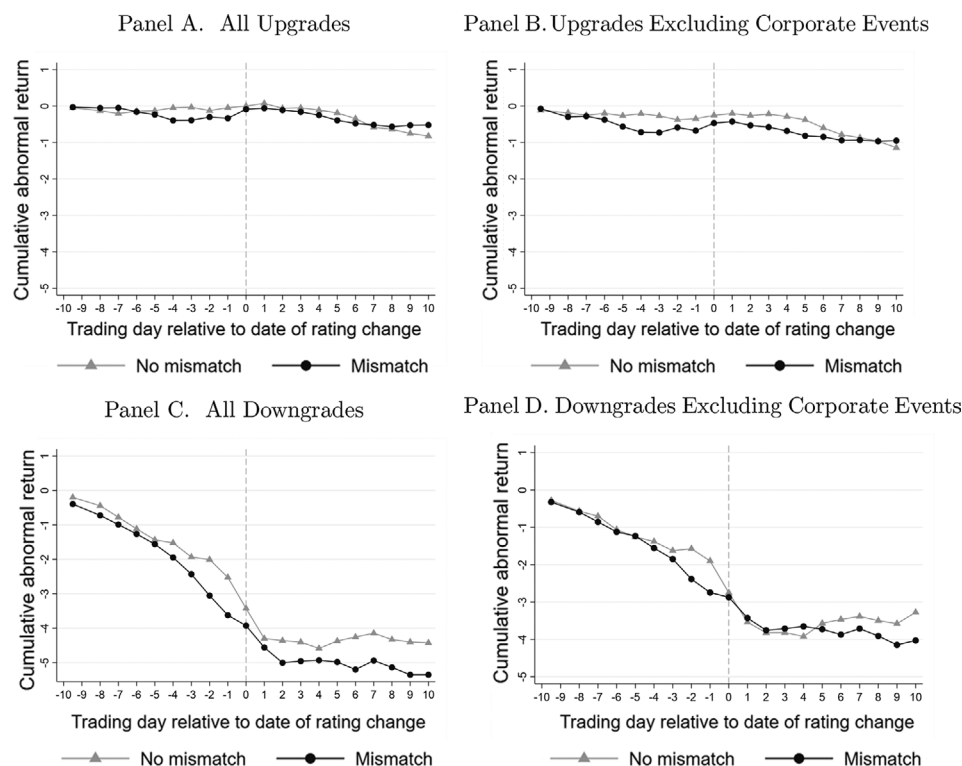


Figure 6. Cumulative abnormal stock returns around rating changes. The figure plots cumulative abnormal stock returns, measured in percent, around rating changes. Panels A and B show returns around upgrades, and Panels C and D show returns around downgrades. Abnormal returns are calculated using the Fama and French (1993) and Carhart (1997) model estimated over trading days $(-300, -50)$ relative to the event date. In Panels B and D, we exclude all rating changes for which a corporate earnings announcement or an M&A announcement falls inside the $(-10, +10)$ window around the rating change.

misaligned analysts? Figure 6 suggests it is not. The overall pattern in the cumulative abnormal returns is very similar for analysts who are aligned and misaligned with the president.²³

Why do stock prices not differentiate more between the downgrades by partisan analysts? Although a complete answer to this question is beyond the scope of this paper, we suggest at least two possible reasons. First, because access to registered voter lists in Illinois, New York City, and New Jersey is limited to the purpose of political campaigns and education, analysts' party affiliation is not public information and therefore may not be fully reflected in the stock price. Second, in the presence of rating-based regulatory frictions, downgrades

²³ Although Figure 6 indicates slightly higher cumulative abnormal returns for misaligned analysts over a three-day window around rating downgrades, this difference disappears when we control for agency and calendar month effects, as we do in Table IA.XIII.

may affect the supply of capital to firms, even if the rating change itself does not reveal any new information to the market.²⁴

Using the cumulative abnormal returns estimated above as a measure of the change in the firm's market capitalization induced by rating changes, we next examine whether firms rated by misaligned analysts experience, on average, a greater reduction in market capitalization. To do so, we aggregate the three-day and seven-day cumulative abnormal returns around rating changes within a given rating agency, firm, and quarter and replace them with zero when no rating change occurs in a given quarter. We then reestimate the regression in equation (1), after replacing the dependent variable with the cumulative abnormal returns around rating changes during the quarter.

Table V presents the results for cumulative abnormal returns measured over two event windows, namely, $(-1,+1)$ (columns (1) and (2)) and $(-3,+3)$ (columns (3) and (4)). In columns (2) and (4), we exclude rating changes with a concurrent corporate event in the event window. We find that in any given quarter after excluding corporate events, firms rated by misaligned analysts lose 3.2 to 3.9 bps more of their market capitalization when they are rated by a misaligned analyst. Because these losses accumulate over time, they are economically sizable: Over a four-year political cycle, a firm rated by a misaligned analyst loses 0.52% ($= 0.0322 \times 16$) to 0.62% ($= 0.0386 \times 16$) more of its market capitalization if it is rated by a misaligned analyst as opposed to an analyst who is politically aligned with the president. For the average-sized firm in our sample, this difference amounts to a total dollar loss of \$89 million to \$107 million.

We believe the estimated losses above represent an underestimation of the true losses in market capitalization due to analysts' partisan perception. First, although voter registration records provide a useful proxy for analysts' political party affiliation, they surely come with measurement error. Classical measurement will bias our estimates of the effect of partisan bias downward. Second, if rating changes are partially anticipated by the market as Figure 6 suggests, the cumulative abnormal returns during the three or seven days around the rating change will underestimate the actual change in the firm's market capitalization due to a rating change. Our estimates above should therefore be interpreted as a lower bound on the magnitude of the stock price effect of analysts' partisan perception.

B. Bond Yields

We next turn to changes in bond yields around rating changes. We obtain daily yield to maturity and yield spreads for bonds issued by our sample of issuers from TRACE (see Section II for a more detailed description of how we process the TRACE data). Yield spreads are computed using the closest-maturity Treasury bond. Because bonds trade less frequently than stocks, we follow Hand, Holthausen, and Leftwich (1992) and compute the

²⁴ See Sangiorgi and Spatt (2017) for an excellent review of the literature on the regulatory role of credit ratings.

Table V
Stock Returns and Bond-Yield Changes around Rating Actions

This table regresses cumulative abnormal stock returns and changes in bond yields around rating changes on *ideological mismatch*. Panel A presents results for cumulative abnormal returns (CARs), which are measured in basis points and calculated using an event window of $(-1,+1)$ (columns (1) and (2)) and $(-3,+3)$ (columns (3) and (4)). Panel B shows results for changes in bond yields ($\Delta Yield$) and bond-yield spreads ($\Delta Yield Spread$) around the rating action, measured in basis points. In quarters with no rating change, the dependent variable is set to zero. In columns (2) and (4) for both panels, we exclude rating changes for which a corporate earnings announcement or M&A announcement falls inside the event window. All variables are defined in the [Appendix](#). *t*-statistics, reported in parentheses, are based on standard errors that allow for double-clustering at the firm-analyst level.

Panel A: Abnormal Stock Returns				
	CAR(-1,+1)		CAR(-3,+3)	
	(1)	(2)	(3)	(4)
Ideological mismatch	-3.2115 (-1.98)	-3.2179 (-2.29)	-4.5770 (-2.04)	-3.8597 (-2.10)
Tenure	-0.3710 (-0.28)	-0.4749 (-0.40)	-1.5297 (-0.76)	-0.7088 (-0.49)
No. of firms covered	0.0347 (0.06)	0.2822 (0.52)	-0.1611 (-0.18)	-0.3161 (-0.42)
Observations	48,147	47,580	48,147	46,934
R^2	0.744	0.750	0.755	0.766
Firm $\times$ Quarter FE	Yes	Yes	Yes	Yes
Agency $\times$ Quarter FE	Yes	Yes	Yes	Yes
Party affiliation FE	Yes	Yes	Yes	Yes
Excluding corporate events	No	Yes	No	Yes
Panel B: Bond Yield Changes				
	$\Delta Yield$		$\Delta Yield Spread$	
	(1)	(2)	(3)	(4)
Ideological mismatch	0.4905 (2.14)	0.4169 (1.90)	0.4575 (2.02)	0.3792 (1.82)
Tenure	0.1141 (1.06)	0.0850 (0.87)	0.1627 (1.56)	0.1361 (1.43)
No. of firms covered	0.0000 (0.00)	0.0048 (0.06)	-0.0764 (-0.90)	-0.0724 (-1.04)
Observations	46,724	46,724	46,724	46,724
R^2	0.699	0.699	0.703	0.694
Firm $\times$ Quarter FE	Yes	Yes	Yes	Yes
Agency $\times$ Quarter FE	Yes	Yes	Yes	Yes
Excluding corporate events	No	Yes	No	Yes
Party affiliation FE	Yes	Yes	Yes	Yes

change in the bond yield (spread) around a rating action as the difference between the bond yield (spread) from the first day following the event and the yield (spread) on the most recent trading day prior to the event. For example, if a bond trades on days -3 , -2 , $+1$, and $+2$, the yield (spread) change is calculated from transaction prices on day -2 and day $+1$. We restrict the sample to bonds that are traded at least once in the 30 days prior and the 30 days after the rating-change announcement. If multiple bonds of the same issuer are traded on the same day, we select a maximum of five bonds with the highest trading volume. In the analysis that follows, yield (spread) changes for each bond for a given firm are equally weighted and the average is treated as a single observation.

First, in Table [IA.XIV](#), we investigate whether yield spreads respond differently to a rating change depending on whether the rating change was initiated by an aligned or misaligned analyst. Consistent with our findings for stock returns, we do not find significant differences in the yield spread reaction. Moreover, Table [V](#), Panel B, mirrors our analysis of abnormal stock returns by examining whether firms rated by misaligned analysts experience on average larger increases in bond yields (spreads) due to rating changes. We first compute the average change in bond yields around the rating actions in our sample.²⁵ Next, we replace the bond yield change with zero in quarters in which no rating change occurs. Finally, we reestimate the regression in equation (1) after replacing the dependent variable with the yield change around rating changes during the quarter. In columns (3) and (4), we repeat the same procedure but use the yield spread change around the rating action rather than the yield change. Our most conservative estimate in column (4) suggests that firms rated by a misaligned analyst experience an increase of 6.1 bps ($= 0.3792 \times 16$) over a four-year term relative to firms rated by aligned analysts.

This point estimate indicates a 29 ($= 6.1/0.21$) bp increase in bond yields for a one-notch rating change. Our estimate is smaller than the estimates from Almeida et al. (2017), who exploit exogenous variation in corporate ratings due to rating agencies' sovereign ceiling policies and find that a 0.7-notch decrease in credit ratings is associated with an increase in bond yields of 34 bps over three months.²⁶ Our estimate is closer to causal estimates from the U.S. municipal bond market: Cornaggia, Cornaggia, and Israelsen (2017) exploit exogenous variation in municipal bond ratings due to a recalibration of Moody's rating scale and find that a one-notch upgrade is associated on average with a decrease in bond yields of 12 to 18 bps.

For the same reasons as discussed above, our estimated effects on bond yields likely represent a lower bound for the true effect of partisan perception. By focusing on a relatively narrow window around the rating change, we

²⁵ In cases in which there are multiple rating actions within a given agency, firm, and quarter, we sum the average change in bond yields around each rating action.

²⁶ A potential explanation for this difference could be the different geographical focus of our study: Whereas we focus on U.S. issuers, Almeida et al. (2017) study an international sample of issuers.

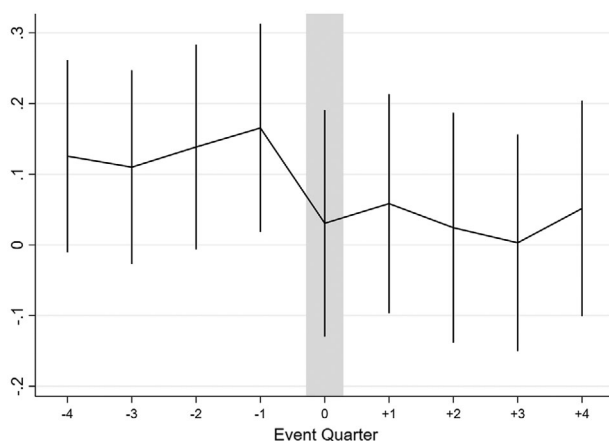


Figure 7. Quarterly event study around presidential elections: Firm investment. The figure plots the difference in log capital expenditures between firms rated by misaligned versus aligned analysts around presidential elections that lead to a change in the party of the president. Specifically, we plot the coefficient estimates for β_t from equation (3). The corresponding 95% confidence intervals are based on standard errors that are clustered at the firm level.

are missing some of the sizable anticipation in bond yields prior to the rating change. In Table IA.XV, we further show that firms rated by misaligned analysts issue new bonds at higher yields than otherwise similar firms rated by aligned analysts. These results are less well identified, because we cannot include firm $\times$ quarter fixed effects as in the analysis above. Instead, we estimate a regression with lagged median rating $\times$ quarter fixed effects and control as best as we can for observable firm characteristics. Unfortunately, we cannot include firm fixed effects as we have very few repeat issuers. In terms of magnitude, the yield spreads of new bonds issued by firms with majority misaligned analysts are 12 bps higher than yields on new bonds issued by firms with majority aligned analysts. These results are useful because they provide direct evidence that firms rated by misaligned analysts experience an increase in their cost of borrowing.

C. Firm Investment

We next test whether partisan perception influences firm investment. In Figure 7, we examine whether firms rated by misaligned analysts experience, on average, a decrease in investment around the election of a new president. As in Figure 3, we focus again on presidential elections in which the party of the president changes (i.e., the 2000, 2008, and 2016 elections) to isolate the effect of a change in analysts' ideological alignment with the president on firm investment. Because investment is observed at the firm-quarter level, we compute a firm-quarter-level measure of *ideological mismatch*, the share of

misaligned analysts covering the firm. Specifically, we estimate the regression:

$$\text{Ln}(\text{Capex}_{f_t}) = \alpha_{fe} + \alpha_t + \sum_{\tau=-4}^{\tau=+4} \beta_{\tau} \text{Mismatch}_{f_t, \tau=0} \times D_t^{\tau} + \epsilon_{f_t}, \quad (3)$$

where $\text{Mismatch}_{f_t, \tau=0}$ is the share of misaligned analysts in a given quarter, with alignment computed based on the party of the president who is elected in $\tau = 0$. To compute the share of misaligned analysts, in each firm-quarter we first compute the share of Democratic (Republican) analysts rating the firm, while treating unaffiliated analysts as in between Democrats and Republicans. Around the 2000 and 2016 elections, we use the share of Democratic analysts as the share of misaligned analysts, and around the 2008 election, we use the share of Republican analysts as the share of misaligned analysts. The dependent variable, $\text{Ln}(\text{Capex})$ is the logarithm of the firm's quarterly capital expenditures. The parameters α_{fe} and α_t correspond to firm $\times$ election event and quarter fixed effects, respectively.

Figure 7 shows a strong decline in capital expenditures around the election for firms rated by analysts who are misaligned with the newly elected president relative to firms rated by aligned analysts. The difference between the average difference in investment levels post election (quarters 0 to +4) and the average difference in investment levels preelection (quarters -1 to -4) is statistically significant at the 5% level (see column (1) of Table Table IA.XVI). In terms of economic magnitude, the decrease in investment by firms that transition from 0% misaligned analysts to 100% misaligned analysts is equal to 11.6%, which corresponds to a decrease of \$19 million for the average firm in our sample. This suggests that firms are reducing capital expenditures when they are rated by misaligned analysts. Moreover, in Table IA.XVI, columns (2) and (3), we show that this result is robust to sorting firms into terciles or quartiles based on their share of misaligned analysts. Figure IA.5 shows that we find a similar pattern if we scale capital expenditures by lagged total assets, with the post-pre election difference in the scaled capital expenditures significant at the 5% level. When we scale by lagged property, plant, and equipment (PPE), we find a statistically significant drop in the investment rate of approximately 0.7 percentage points during the election quarter. After the election quarter, the difference in investment rate is not significantly different from zero. These results suggest that companies rated by misaligned analysts adjust almost immediately to a lower level of capital stock following the election. Once the denominator, that is, lagged capital, equals the new steady state, the estimated effect of the election on the investment rate is driven towards zero by construction (see Yagan (2015)).

A potential concern about the analysis above is that firms rated by aligned versus misaligned analysts could differ on unobserved dimensions, which could lead to a differential response to a presidential election even in the absence of differential rating actions. This concern is difficult to address, because it would require exogeneity in the assignment of analysts to firms. Although the analysis of firm investment has more shortcomings compared to

our previous tests, which include firm $\times$ quarter fixed effects, we nevertheless find the analysis useful because it provides strong suggestive evidence that the differences in analysts' rating behavior documented above have real consequences. This finding is also in line with prior evidence in the literature that credit ratings affect firm investment (e.g., Almeida et al. (2017)).

D. Analyst Accuracy

Our main results suggest that analysts make different rating adjustments for the same firm at the same point in time depending on whether the White House is controlled by the party with which they are affiliated. Because, by construction, these rating differences are orthogonal to firm fundamentals, we would expect partisan perceptions to also affect credit rating accuracy. However, based on existing results, we cannot disentangle which of the two groups of analysts—those with and without *ideological mismatch*—is relatively more “right” or “wrong.” In other words, whether the analysts who are aligned with the president's party downward-adjust by too little, whether those who are misaligned downward-adjust by too much, or whether both of these effects obtain is unclear.

Finding suitable measures of analyst accuracy is challenging for a number of reasons. First, no clear benchmark of analysts who are not subject to partisan perception exists. Using unaffiliated or unregistered analysts as a “neutral” benchmark might be tempting, but both groups suffer from substantial measurement error. In particular, given the location of the analysts in our sample (Chicago and New York), the chance that an unaffiliated or unregistered analyst is Democratic leaning is high, which would bias us toward finding that Democratic analysts are less biased. Second, the firm's true probability of default and expected loss given default are unobservable, and traditional metrics of rating accuracy that are based on ex post default rates by rating category rely on a large number of sample events in order to be meaningful. Third, due to the causal effect of credit ratings (see Almeida et al. (2017)), ex post firm outcomes reflect both firm fundamentals in the absence of credit ratings and the causal effect of the credit rating. Fourth, determining systematic differences across analysts in their ex post rating accuracy across Democratic and Republican presidents requires a long time series of election cycles.

To at least partially address these challenges, we present two distinct tests of analyst accuracy. To rule out that our coding of unaffiliated analysts influences the results, we restrict our analysis below to Democratic and Republican analysts. Our first measure builds on Fracassi, Petry, and Tate (2016) and Kisgen et al. (2020) by capturing the correlation between rating changes and future movements in yield spreads. Intuitively, if the analyst issues a downgrade ($\Delta R_{ift} > 0$) and credit spreads on the firm's bonds subsequently increase (decrease), she would be considered more accurate (inaccurate). We operationalize this idea by regressing future changes in yield spreads ($s_{f,t+h} - s_{ft}$) on the analyst's current-quarter change in the rating (ΔR_{ift}),

ideological mismatch (M_{it}), an interaction term, and controls,

$$s_{f,t+h} - s_{ft} = \alpha_{jt} + \alpha_p + \beta_1 \Delta R_{ift} + \beta_2 \Delta R_{ift} \times M_{it} + \beta_3 M_{it} + \gamma' X_{ft} + \epsilon_{ift}. \quad (4)$$

We control for agency $\times$ quarter and party affiliation fixed effects as in our baseline regression. Because we are unable to include firm $\times$ quarter fixed effects as yield spreads do not vary within firm-quarter, we include median lagged rating $\times$ quarter fixed effects as well as controls for observable firm characteristics (X_{ft}). A positive (negative) coefficient β_2 in equation (4) would indicate higher (lower) accuracy for mismatched analysts.

Table VI, Panel A, reports the results. We vary the horizon over which the future change in yield spreads is measured from four to eight to 12 quarters. Across all horizons, we find the interaction between the current-quarter rating change and *ideological mismatch* is negative, indicating lower accuracy for misaligned analysts. Taken together, the results imply that rating changes by ideologically mismatched analysts are less positively correlated with future changes in spreads.

An important drawback of spread-based accuracy measures is that credit spreads may be directly affected by analysts' rating actions. In particular, Almeida et al. (2017) show that bond yields continue to increase for several months following a downgrade. To mitigate this problem to some degree, our second measure builds on existing work by Kempf (2020) and Jiang, Kumar, and Law (2016) by measuring analyst (in)accuracy based on future revisions of the previously assigned rating. Specifically, we measure accuracy as -1 times the absolute difference between the current rating and the prevailing rating four, eight, or 12 quarters in the future. The intuition behind this measure is that larger revisions in the future indicate less accurate ratings.

Table VI, Panel B, reports the results from regressing this measure of accuracy on analysts' *ideological mismatch*. All specifications include agency $\times$ quarter fixed effects, firm $\times$ quarter fixed effects, and party affiliation fixed effects. Across all horizons, we observe that ideologically mismatched analysts are less accurate, as their ratings need to be revised more in the future. The point estimate in column (3) indicates that *ideological mismatch* reduces analyst accuracy by 0.102 notches, which corresponds to 11.2% of the mean in analyst accuracy measured over 12 quarters. These results therefore confirm the conclusion from our first test that misaligned analysts are less accurate.

To summarize, the results in Table VI suggest that analysts who do not support the president tend to be less accurate. Measuring rating accuracy using yield spreads and deviations between the initial rating and subsequent ratings has important advantages. First and foremost, it does not require a large number of ratings for each individual analyst to be meaningful. Second, it allows us to capture smaller changes in the expected default probability that may not always lead to a default. Third, to the extent that future rating revisions or changes in spreads are driven by the arrival of new information, this information would affect the accuracy measure of all analysts rating the same firm at the same point in time.

Table VI
Analyst Accuracy

This table compares the rating accuracy of analysts with and without *ideological mismatch*. In Panel A, the change in yield spreads over the following four, eight, and twelve quarters is regressed on the current-quarter rating change, *ideological mismatch*, as well as an interaction term. Firm controls include lagged firm size, Tobin's Q, investment, leverage, cash holdings, and cash flow. In Panel B, analyst accuracy, defined as -1 times the absolute change between the current rating and the rating four, eight, and 12 quarters in the future, is regressed on *ideological mismatch*. The sample is restricted to Democratic and Republican analysts in both panels. *t*-statistics, reported in parentheses, are based on standard errors that allow for double-clustering at the firm-analyst level.

Panel A: Accuracy Based on Future Changes in Yield Spreads			
	Δ Yield Spread		
	4Q (1)	8Q (2)	12Q (3)
Ideological mismatch	−0.0686 (−1.08)	−0.0562 (−0.75)	0.0130 (0.16)
Mismatch $\times$ Rating change	−0.2771 (−2.17)	−0.2922 (−2.46)	−0.1948 (−1.75)
Rating change	−0.0011 (−0.01)	−0.0979 (−0.93)	−0.0855 (−0.81)
Observations	15,538	13,574	11,715
R^2	0.656	0.664	0.644
Agency $\times$ Quarter FE	Yes	Yes	Yes
Median lagged rating $\times$ Quarter FE	Yes	Yes	Yes
Party affiliation FE	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes
Panel B: Accuracy Based on Future Rating Revisions			
	Accuracy		
	4Q (1)	8Q (2)	12Q (3)
Ideological mismatch	−0.0406 (−1.99)	−0.0879 (−2.49)	−0.1021 (−2.64)
Tenure	−0.0064 (−0.63)	−0.0191 (−0.94)	−0.0100 (−0.35)
No. of firms covered	0.0076 (1.25)	0.0199 (1.67)	0.0164 (0.96)
Observations	25,100	20,672	16,961
R^2	0.872	0.891	0.897
Firm $\times$ Quarter FE	Yes	Yes	Yes
Agency $\times$ Quarter FE	Yes	Yes	Yes
Party affiliation FE	Yes	Yes	Yes

Although these tests allow us to address some of the empirical challenges associated with establishing differences in accuracy that we highlighted above, some important caveats remain. First, determining systematic differences across analysts in their ex post rating accuracy under Democratic and Republican presidents with sufficient confidence requires a much longer time series of election cycles than our current sample period. Second, and perhaps more importantly, any test of analyst accuracy involves a comparison of ex ante credit ratings with ex post outcomes (e.g., defaults, rating revisions, or credit spreads). Due to the causal effect of credit ratings, ex post outcomes reflect both firm fundamentals in the absence of the credit rating and the causal effect of the rating. Hence, one of the crucial advantages of our setting—that credit ratings having causal effects on securities prices and firm policies—is also a drawback when testing analyst accuracy.

VI. Mechanisms

Thus far we document that partisan perception affects credit ratings and has both price and real effects for firms. In this section, we investigate potential channels through which political party affiliation influences credit ratings.

A. *Belief Disagreement*

Existing literature establishes a strong influence of political alignment with the U.S. president on households' beliefs about aggregate economic conditions (e.g., Bartels (2002), Gerber and Huber (2009), Mian, Sufi, and Khoshkhoh (2018)). Thus, a natural explanation for our results could be that analysts with different party affiliations differ in their views on how the economic policies and regulations of the U.S. president will affect the credit risk of firms in the economy.

We provide three pieces of evidence to support the interpretation that our results are driven by belief disagreement. First, we conduct an online survey of credit rating analysts and document more pessimistic views on current economic conditions among credit rating analysts who are not aligned with the president. Second, we find that the effect is stronger when political polarization in views of the economic condition is high. Third, we show that differences in rating behavior disappear when analysts rate firms whose fundamentals are less correlated with the U.S. economy and with presidential policies.

A.1. *Survey Evidence*

We first present survey evidence to directly support our interpretation that credit analysts with different partisan identities differ in their economic beliefs. To keep the survey as brief as possible and maximize the potential response rate, we focus on the state of the economy as the object of analyst disagreement. Specifically, we conducted an online survey of current and former credit rating analysts to ask them about their views on current

economic conditions as well as about their political ideology and demographic characteristics. We invited alumni of Cornell University's Johnson School of Management and of the University of Chicago's Booth School of Business who work or used to work at credit rating agencies, as well as members of the LinkedIn group "Credit Rating Agency Alumni + Friends," to participate in the survey. Overall, we invited 402 individuals and received 91 responses (i.e., we have a response rate of 22.6%).

To infer analysts' views on current economic conditions, we used the following question from the Gallup Daily survey: "How would you rate economic conditions in this country today?" Possible responses were "excellent," "good," "fair," and "poor." We code the answers with numerical values from 1 to 4, with 1 indicating the least favorable view and higher values representing more favorable views. In addition to the question on economic views, we also asked analysts about their number of years of experience and their demographic characteristics, such as age, gender, ethnicity, and residence. To infer the political affiliation of the analysts, we use the question from the Gallup Daily survey: "In politics, as of today, do you consider yourself a Republican, a Democrat, an Independent or Unaffiliated?" We classify a respondent's political affiliation as *Republican*, *Democrat*, or *independent* (if she responded Independent or Unaffiliated). We drop 17 individuals who indicated they had never worked as credit rating analysts and 17 respondents who did not finish the survey. Our final sample of individuals who responded to both questions consists of 57 responses.

Table [IA.II](#) presents summary statistics on the respondents. The vast majority of the respondents had more than 10 years of work experience as credit rating analysts, with 35% of respondents having more than 15 years of experience. Regarding their political affiliation, 39% identified as Democrat, 28% as Republican, and 33% as Independent, which is very similar to the political composition in our main sample. Also consistent with our main data set, the majority of the respondents were white and male.

Table [VII](#) presents results on analysts' self-reported views on the state of the economy. We find that Republicans held a more positive view of current economic conditions than Independents, whereas Democrats were statistically indistinguishable from Independents. The difference is robust to including controls for analysts' demographic characteristics (gender, age, and ethnicity) and years of experience. The difference between Democrats and Republicans is statistically significant at the 1% level in column (1) and at the 10% level in columns (2) and (3), and ranges between 0.366 (column (2)) and 0.432 (column (1)). Because the survey was conducted under a Republican administration, the results confirm the findings from prior household surveys that individuals' views of economic conditions depend on whether the White House is occupied by their preferred political party (Bartels (2002), Gaines et al. (2007), Gerber and Huber (2009), Curtin (2016), Mian, Sufi, and Khoshkhrou (2018)). The economic magnitude of the difference between Democrats and Republican is also plausible. As of the fourth quarter of 2017, the last quarter with available data from the Gallup Daily survey, the difference in views of economic conditions between Republican and Democratic households was 0.57. The difference

Table VII
Survey of Credit Rating Analysts

This table shows results from our online survey of credit rating analysts. We infer analysts' views on current economic conditions by asking the question: "How would you rate economic conditions in this country today?", with possible answers of "excellent," "good," "fair," and "poor." We code the answers with values from 1 to 4, with 1 indicating the least favorable view and higher values representing more favorable views. To infer the political affiliation of the analysts, we ask, "In politics, as of today, do you consider yourself a Republican, a Democrat, an Independent or Unaffiliated?" We classify a respondent's political affiliation as Republican, Democrat, or independent (if she responded Independent or Unaffiliated). *Demographic controls* include gender, age, and ethnicity. *Experience controls* is the self-reported number of years of work experience as a credit rating analyst. *t*-statistics, reported in parentheses, are based on robust standard errors.

	Current Economic Conditions		
	(1)	(2)	(3)
Democrat	0.1866 (1.19)	0.1903 (1.16)	0.1835 (1.11)
Republican	0.6184 (3.89)	0.5559 (3.14)	0.5639 (3.13)
Constant	2.6316 (23.15)	2.8109 (11.19)	2.8486 (10.66)
Observations	57	54	54
R^2	0.214	0.285	0.287
Demographic controls	No	Yes	Yes
Experience controls	No	No	Yes

in our sample of credit rating analysts is somewhat smaller, but it is still economically large, which is remarkable given that we are studying a sample of financial experts who have a much higher level of economic sophistication than the average U.S. household. Moreover, the result provides direct evidence for our interpretation that the differences in credit rating actions documented above could be driven by differences in economic beliefs.

A.2. Political Polarization in Views of Economic Conditions

To further support the interpretation that our results reflect differences in economic beliefs, we investigate whether the effect of partisan bias is more pronounced during periods in which views of economic conditions are more politically polarized in the U.S. population.

To capture political polarization in the views of economic conditions, we use data from the Gallup Daily survey. To capture the views on current economic conditions, Gallup asks the question: "How would you rate economic conditions in this country today — as excellent, good, only fair, or poor?" Responses to this question are converted into a scale that ranges from 1 (poor) to 4 (excellent). Our measure of polarization is the absolute difference in the average economic views between Democrats and Republicans in a given calendar quarter. We

Table VIII
Interaction with Polarization of Economic Views

This table regresses quarterly rating changes on *ideological mismatch* as well as on interactions with a measure of political polarization of economic views (*Econ. polarization Gallup*). To measure political polarization in the views of economic conditions, we use data from the following question in the Gallup Daily survey: “How would you rate economic conditions in this country today—as excellent, good, only fair, or poor?” Responses to this question are converted into a numerical scale that ranges from 1 (poor) to 4 (excellent). Based on this question, our measure of polarization of economic views, *Econ. polarization Gallup*, is the absolute average quarterly difference between Democrats and Republicans. We standardize the measure to have a mean of zero and a standard deviation of one. *t*-statistics, reported in parentheses, are based on standard errors that allow for double-clustering at the firm-analyst level.

	Rating Change		
	(1)	(2)	(3)
Ideological mismatch	0.0105 (2.02)	0.0109 (2.10)	0.0085 (1.89)
Mismatch × Econ. polarization Gallup	0.0087 (2.96)	0.0084 (2.82)	0.0053 (1.93)
Tenure	−0.0004 (−0.15)	−0.0011 (−0.40)	−0.0003 (−0.12)
No. of firms covered	−0.0001 (−0.08)	0.0005 (0.39)	0.0003 (0.22)
Observations	45,041	45,041	45,041
<i>R</i> ²	0.805	0.806	0.808
Firm × Quarter FE	Yes	Yes	Yes
Agency FE	Yes	No	No
Agency × Sector FE	No	Yes	No
Agency × Quarter FE	No	No	Yes
Party affiliation FE	Yes	Yes	Yes

standardize the measure to have a mean of zero and a standard deviation of one.

In Table VIII, we interact *Ideological mismatch* with the polarization in economic views *Econ. polarization Gallup*. The estimates in column (1) imply that the effect of misalignment with the president increases by 0.011 notches for a one-standard-deviation increase in polarization, thereby increasing the baseline effect by 83% (= 0.0087/0.0105). The result is robust to adding agency × sector fixed effects (column (2)) and agency × quarter fixed effects (column (3)). Table IA.XVII shows that we find similar results if we construct our measure of polarization in economic views based on the Michigan Survey.

A.3. Firm Heterogeneity

If the main result in this paper is driven by credit analysts with different party affiliations having different views on how the economic policies of the U.S. president affect the credit risk of firms in the economy, we should see

Table IX
Firm Heterogeneity

This table regresses quarterly rating changes on *ideological mismatch* as well as interactions with two measures of firm cyclicality. *Beta* refers to the firm-level market beta and is estimated using the Fama and French (1993) and Carhart (1997) four-factor model, monthly return data from CRSP, and a five-year rolling window. *Sector avg. beta* is computed as the average *Beta* of all firms in a given GICS sector and quarter. *t*-statistics, reported in parentheses, are based on standard errors that allow for double-clustering at the firm-analyst level.

	Rating Change			
	(1)	(2)	(3)	(4)
Ideological mismatch	−0.0013 (−0.18)	−0.0023 (−0.33)	−0.0091 (−0.85)	−0.0078 (−0.77)
Mismatch × Beta	0.0130 (2.62)	0.0115 (2.35)		
Mismatch × Sector avg. beta			0.0227 (2.31)	0.0188 (2.10)
Tenure	−0.0018 (−0.60)	−0.0011 (−0.37)	0.0002 (0.08)	−0.0003 (−0.11)
No. of firms covered	0.0000 (0.02)	−0.0002 (−0.18)	−0.0002 (−0.14)	−0.0001 (−0.08)
Observations	33,745	33,739	49,792	49,790
R^2	0.789	0.794	0.804	0.808
Firm × Quarter FE	Yes	Yes	Yes	Yes
Agency FE	Yes	No	Yes	No
Agency × Quarter FE	No	Yes	No	Yes
Party affiliation FE	Yes	Yes	Yes	Yes

smaller differences in rating behavior when analysts rate firms whose fundamentals are less correlated with the U.S. economy. To measure cyclicality, we compute firms' equity market betas using the Fama and French (1993) and Carhart (1997) model estimated on monthly returns and a five-year rolling window. We then interact *Ideological mismatch* with the estimated market beta. Table IX, columns (1) and (2), report the results. We find that the partisan disagreement is substantially more pronounced for firms with high market betas, that is, high cyclicality. We obtain a similar result if we compute the average market beta within a given GICS sector and quarter and then interact *ideological mismatch* with the sector-level market beta (columns (3) and (4)). The effect of partisan perception thus appears to be concentrated in firms with high exposure to the U.S. economy.

B. Alternative Mechanisms

Although the analysis above strongly supports belief disagreement as a potential mechanism driving our results, other nonmutually exclusive mechanisms could be present. Although we cannot exclude all potential

alternative explanations, our evidence so far allows us to rule out several alternative stories.

Analysts' Personal Economic Situation. An important advantage of our setting is that rating actions are unlikely to be driven by differences in how analysts' personal economic situation changes when their preferred party is in power. This confounding factor represents a key challenge for studies of household behavior, but how analysts' professional assessment of the credit risk of firms in the economy could be affected by their personal economic situation is not obvious. Moreover, because the analysts whom we are comparing work for the same rating agency at the same point in time, their income is unlikely to be differentially affected by the elected president. Hence, the concern that our results could reflect differences in the wealth or income of the individuals whom we study is mitigated in our setting.

Strategic Behavior. Prior literature shows that equity analysts may be biased due to strategic behavior (e.g., Laster, Bennett, and Geoun (1999), Hong and Kostovetsky (2012)). These studies raise the question of whether our documented effect reflects strategic behavior rather than belief disagreement.

We believe that our empirical setting substantially mitigates this concern. For strategic behavior to affect our estimates, it would have to not only correlate with political affiliation, but also vary with the party of the president. We feel it is nontrivial to think of plausible stories that could generate such strategic behavior.

One potential strategic behavior that would vary with the president's party could be that misaligned analysts downgrade firms more frequently in an effort to harm the economy and punish the president's party. Although this explanation is theoretically consistent with many results that we document, we find it less plausible for two main reasons. First, it assumes a high degree of coordination among analysts, because the rating decisions of any individual corporate analyst are unlikely to materially affect the economy. Such coordination would be especially difficult given that each analyst has an incentive to deviate, as her future promotion and career outcomes may depend on how accurately she assesses credit risk. Second, results from the event study around the 2016 election are more consistent with belief disagreement and less consistent with misaligned analysts attempting to harm the president. If analysts were trying to harm the president, we would have expected no divergence in rating actions in response to the election outcome and a larger divergence when Donald J. Trump took office. Instead, we find that Democratic analysts downward-adjusted credit ratings more relative to Republican analysts during the month when the election result became public, with no significant difference in the month when President Trump took office.

Another source of strategic behavior could be related to analysts' career concerns. For example, analysts could be rewarded with a career in the public sector in return for expressing positive views when their own party is in power, and negative views when the other party is in power. We think this story is unlikely to explain our results for two reasons. First, more direct and visible ways to support a political party are available than issuing optimistic or

pessimistic corporate bond ratings. For example, sovereign credit ratings would likely be more salient to a party's administration. Second, as an empirical fact, corporate credit analysts rarely pursue a career in the public sector. We hand-collected information on the career paths of the analysts in our sample and find that only five out of 557 analysts started a public-sector job following their employment at the agency.

Political Connections. Another potential explanation is that analysts who are affiliated with the president's party may have "inside" information due to political connections. We think political connections are unlikely to explain our results for at least three reasons. First, inside information due to political connections would predict a difference in rating accuracy, but not necessarily a difference in the direction of the average rating change, because it is not obvious why the inside information that analysts receive from their sources would always be more positive relative to the beliefs of other analysts. Second, this story is inconsistent with the timing of the effect around the 2016 presidential elections documented above. If the results reflected political connections, we should start to see a divergence shortly after Trump was inaugurated. Instead, we see a sizable divergence during the month of the election and no significant divergence in the month when he takes office. Third, we hand-collected information on the career paths of the analysts in our sample and found that only three analysts ended up transitioning to a job in the federal government after their employment at the rating agency. These numbers suggest a relatively low degree of political connections in our sample of analysts.

To conclude, although there is a large literature on the consequences of investor disagreement on asset prices, we know relatively little empirically about the sources of disagreement or factors that can intensify disagreement (e.g., Cookson and Niessner (2020)). Our empirical evidence establishes political partisanship as an important source of disagreement. Compared to other studies that infer differences in beliefs from differences in trading behavior, the strength of our empirical setup is that we can rule out other important drivers of dispersion in trading decisions, such as differences in hedging demands or liquidity needs as well as differences in private information. Moreover, by complementing our analysis of rating actions with the survey, we provide direct evidence that the differences in rating actions we observe are likely due to differences in economic beliefs. Another important insight from our study is that analyst beliefs do not converge over the course of a presidency, which is consistent with frictions in learning (Xiong (2013)).

VII. Conclusion

We show that partisan perception affects the decisions of financial analysts. Using a novel data set that links credit rating analysts to party affiliations from voter registration records, we show that analysts who are not aligned with the president's party are more likely to adjust ratings downward. Our identification approach compares analysts with different party affiliations covering the same firm at the same point in time, ensuring that differences in

firm fundamentals cannot explain the observed differences in rating actions. Moreover, we provide evidence that the difference in rating actions is likely driven by differences in beliefs about aggregate economic conditions. Our results therefore suggest that analysts have partisan models of the world and update their beliefs heterogeneously over the political cycle. Finally, we show that rating actions by partisan analysts have significant price effects and may influence firms' investment decisions.

What could be the source of the partisan perception documented in this paper? Although a complete answer to this question is beyond the scope of our paper, we speculate that partisan perception could reflect two factors. First, it could reflect a causal effect of party affiliation whereby analysts, similar to sports fans, feel optimism or pessimism because their team won the election (Mason (2015)). Alternatively, party affiliation could be a powerful statistic that summarizes ideological differences across analysts beyond ethnicity, gender, and age and has strong incremental predictive power for their rating decisions and their response to public signals.

Given the documented increase in political polarization, understanding the potential implications of this trend for the U.S. economy is important. One potential channel for how polarization can have real effects is through the economic beliefs and actions of relevant economic agents. To the best of our knowledge, this study is the first to quantify the influence of partisan perception on the actions of an important set of finance professionals (credit rating analysts), as well as on securities prices and firm investment. Because the effect of partisan perception prevails even in a setting where pecuniary and professional gains are at stake, it may influence the behavior of many other relevant economic agents, such as firm managers and asset managers. This avenue is thus a fruitful one for future research.

Appendix: Variable Definitions

Variable	Description
<i>Dependent variables</i>	
Rating change	The quarterly change (measured in notches) in the credit rating of a given firm by a given rating agency. Credit ratings are transformed into a cardinal scale, as in Fracassi, Petry, and Tate (2016), starting with 1 for AAA and ending with 21 for D or lower for S&P and Fitch. For Moody's, the scale starts with 1 for Aaa and ends with 21 for C. Credit ratings are obtained for S&P from S&P RatingXpress, for Moody's from Moody's Default and Recovery Database, and for Fitch from Mergent FISD. The variable is winsorized at the top and bottom 1%.
CAR(−1,+1); CAR(−3,+3) around rating changes	Cumulative abnormal returns (CARs) during trading days (−1,+1) and (−3,+3) around a rating change, computed using the Fama and French (1993) and Carhart (1997) model estimated over trading days (−300, −50) relative to the event date. CARs are aggregated over all rating changes for a given agency, firm, and quarter, and are set to zero when no rating change occurs. We exclude rating actions with missing stock returns in the (−3,+3) window around the event. The variable is measured in basis points and is winsorized at the top and bottom 1%.
Δ Yield (spread) around rating changes	The change in bond yields (spreads) around a rating change, averaged across all outstanding bonds by the issuer. Bond yield (spread) changes around a rating action are computed as the difference between the bond yield (spread) on the first day following the event and the yield (spread) on the most recent trading day prior to the event. Average bond yield (spread) changes are then aggregated over all rating changes for a given agency, firm, and quarter, and are set to zero when no rating change occurs. We restrict the sample to bonds that are traded at least once in the 30 days prior and the 30 days after the rating change announcement. If multiple bonds of the same issuer are traded on the same day, we select a maximum of five bonds with the highest trading volume. The benchmark rate that is used to compute credit spreads is based on interpolation of the yields of the two government bonds with the next-lower and the next-higher duration relative to the corporate bond. The variable is measured in basis points and is winsorized at the top and bottom 1%.
Accuracy over 4, 8, and 12 quarters	−1 times the absolute difference between the current rating and the prevailing rating four, eight, or 12 quarters in the future. The variable is measured in notches.
Δ Yield spread over 4, 8, and 12 quarters	The difference in the yield spread between quarter t and quarter four, eight, and 12 in the future. We compute yield spreads as the difference between a bond's yield to maturity on a given trading day and the benchmark rate, winsorized at the top and bottom 1%. The benchmark rate is based on interpolation of the yields of the two government bonds with the next-lower and the next-higher duration relative to the corporate bond. For each bond we keep the yield to maturity as of the last trading day of the quarter and then average across all outstanding bonds by the same issuer. If multiple bonds of the same issuer are traded on the same day, we select a maximum of five bonds with the highest trading volume.

Variable	Description
<i>Main independent variables</i>	
Ideological mismatch	Indicator function equal to one if the analyst's party affiliation does not match the party of the president in a given quarter, and zero if the party matches or the analyst is unaffiliated. Information on party affiliation is obtained after merging analysts to voter records from Illinois, New Jersey, and New York City. Section I of the Internet Appendix provides additional details regarding the voter files and the merging procedure.
<i>Control variables</i>	
Tenure	Logarithm of one plus the number of quarters since the analyst's first rating action for a given rating agency.
No. of firms covered	Logarithm of the number of firms rated by the analyst in a given quarter.
Votes in midterm or primary	An indicator equal to one for analysts who have voted in midterm or primary elections, and zero otherwise.
Avg. election gap	Log average time gap (in quarters) between elections in which the analyst has voted.
Size	The lagged logarithm of the firm's total assets from Compustat.
Investment	The lagged ratio of the firm's capital expenditures to the lagged value of property, plant, and equipment from Compustat.
Leverage	The lagged ratio of the firm's total long-term debt to total assets from Compustat.
Tobin's Q	The lagged ratio of the firm's quarterly market value to book value of total assets from Compustat. It is winsorized at the top and bottom 1%.
Cash	The lagged ratio of the firm's cash and short-term investments to total assets from Compustat.
Cash flow	The lagged ratio of the firm's income before extraordinary items and depreciation to lagged property, plant, and equipment from Compustat.
Median lag rating	The median lagged rating across all rating agencies rating the firm.
Econ. polarization Gallup	The absolute difference between the average economic views of Democrats and Republicans in a given quarter from Gallup Daily survey.
Beta	Equity market beta estimated using the Fama and French (1993) and Carhart (1997) four-factor model, using monthly return data from CRSP and a five-year rolling window.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.